Q1

Find the increase in pressure required to decrease the volume of water sample by 0.01%. Bulk modulus of water $= 2.1 \times 10^9\ Nm^{-2}$.

(1) $4.3 \times 10^4\ N/m^2$

(2) $1.8 \times 10^7 N/m^2$

(3) $2.1 \times 10^5 N/m^2$

(4) $3.7 \times 10^4\ N/m^2$

Q2

Assertion: Magnetic moment of an atom is due to both, the orbital motion and spin motion of every electron.

Reason: A charged particle produces a magnetic field.

(1) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(2) If both Assertion and Reason are true but Reason is not explanation of the Assertion.

(3) If Assertion is true but the Reason is false.

(4) If Assertion is false but Reason is true.

Q3

Two identical cells connected in series send $1.0\ A$ current through a $5\ \Omega$ resistor. When they are connected in parallel, they send $0.8\ A$ current through the same resistor. What is the internal resistance of the cell?

(1) $0.5\ \Omega$

(2) $1.0\ \Omega$

(3) $1.5\ \Omega$

(4) $2.5\ \Omega$

Q4

Which of the following set have different dimensions?

(1) Pressure, Young's modulus and stress.

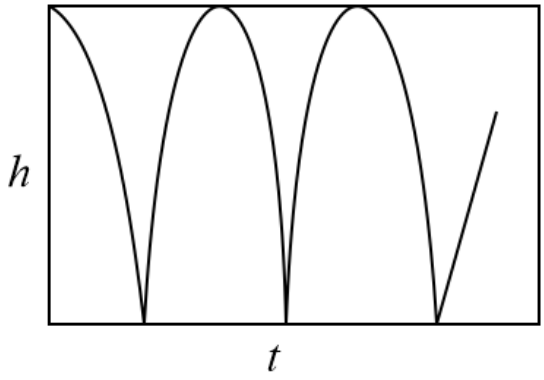
(2) E.M.F., potential difference and electric potential.

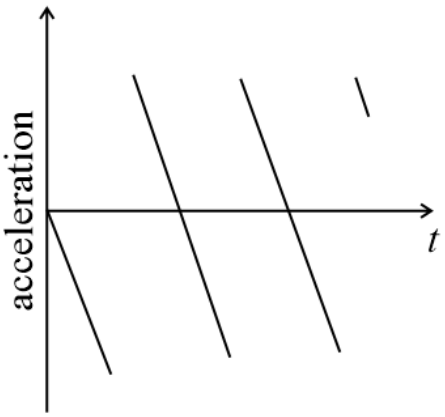
(3) Heat, work done and energy.

(4) Dipole moment, electric flux and electric field.

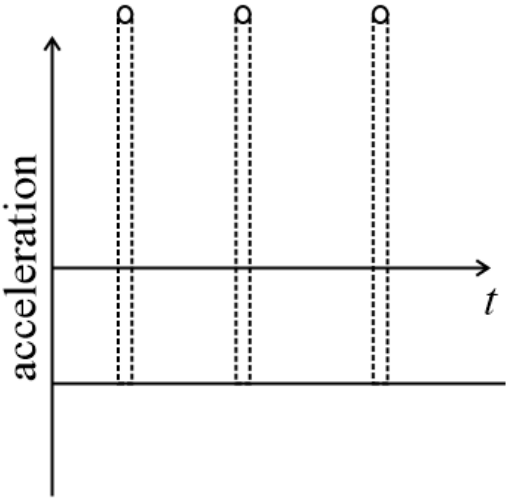
Q5

A ball is dropped vertically from height h and is bouncing elastically on the floor (see figure). Which of the following plots best depicts the acceleration of the ball as a function of time.

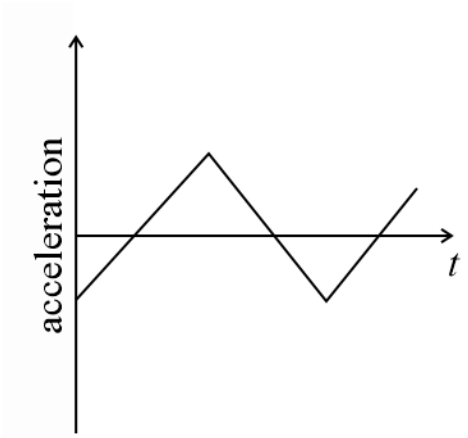




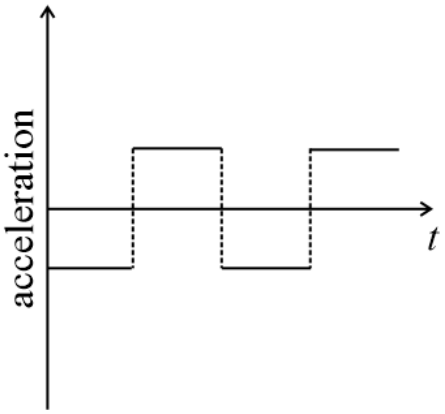
(1)



(2)



(3)



(4)

Q6

A body is sliding down an inclined plane (angle of inclination 45°). If the coefficient of friction is 0.5 and $g = 9.8\ \mathrm{m\ s^{-2}}$. then the downward acceleration of the body in $\mathrm{m\ s^{-2}}$ is :-

(1) $\frac{4.9}{\sqrt{2}}$

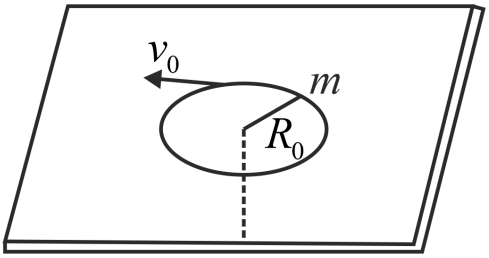
(2) $4.9\sqrt{2}$

(3) $19.6\sqrt{2}$

(4) 4.9

Q7

A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0 . The mass is attached to a string which passes through a smooth hole in plane as shown.



The tension in the string is increased gradually and finally m moves in a circle of radius $\frac{R_0}{2}$. The final value of the kinetic energy is:

(1) mv_0^2

(2) $\frac{1}{4}mv_0^2$

(3) $2\ mv_0^2$

(4) $\frac{1}{2}mv_0^2$

Q8

A charged oil drop is suspended in a uniform field of $3 \times 10^4\ \mathrm{V/m}$ so that it neither falls nor rises. The charge on the drop will be (take the mass of the charge = $9.9 \times 10^{-15}\ \mathrm{kg}$ and $g = 10\ \mathrm{m/s^2}$)

(1) $3.3 \times 10^{-18}\ \mathrm{C}$

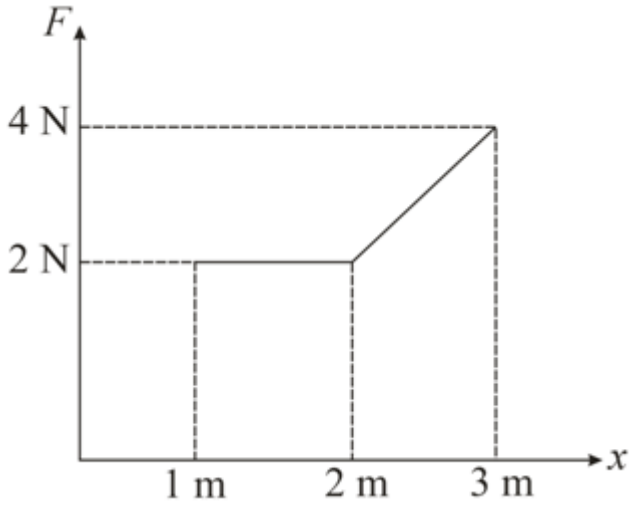
(2) 3.2×10^{-18} C

(3) 1.6×10^{-18} C

(4) 4.8×10^{-18} C

Q9

Starting from rest, a particle is moving along x-axis. The variation of force with position is shown in figure. Find the Kinetic energy (KE) of particle when it is at $x = 3\text{m}$.



(1) 3.3 J

(2) 4 J

(3) 5 J

(4) 4.2 J

Q10

A planet is revolving around the Sun in an elliptical orbit. Its closest and farthest distance from the Sun are r_{min} and r_{max}, respectively. If the orbital angular velocity of the planet when it is nearest to the Sun is ω, then the orbital angular velocity at the point when it is at the farthest distance from the Sun is

(1) $\sqrt{\frac{r_{\text{min}}}{r_{\text{max}}}}\omega$

(2) $\sqrt{\frac{r_{\text{max}}}{r_{\text{min}}}}\omega$

(3) $\left(\frac{r_{\text{max}}}{r_{\text{min}}}\right)^2\omega$

(4) $\left(\frac{r_{\min}}{r_{\max}}\right)^2 \omega$

Q11

Find the time required for a $50\ \text{Hz}$ alternating current to change its value from zero to the rms value.

(1) $10\ \text{ms}$

(2) $5\ \ \text{ms}$

(3) $15\ \text{ms}$

(4) $2.5\ \text{ms}$

Q12

If two waves of the same frequency and amplitude respectively on superposition produce a resultant disturbance of the same amplitude, the waves differ in phase by

(1) Π

(2) Zero

(3) Π/3

(4) 2π/3

Q13

A radiowave has a maximum magnetic field induction of $10^{-4}\ \text{T}$ on arrival at a receiving antenna. The maximum electric field intensity of such a wave is

(1) zero

(2) $3 \times 10^4\ \text{V m}^{-1}$

(3) $5.8 \times 10^{-4}\ \text{T}$

(4) $3.0 \times 10^{-5}\ \text{T}$

Q14

Choose the correct statement for hydrogen and deuterium atoms (considering the motion of nucleus).

(1) The radius of first Bohr orbit of deuterium is less than that of hydrogen.

(2) The speed of electron in first Balmer line of deuterium is more than that of hydrogen.

(3) The wavelength of first Balmer line of deuterium is more than that of hydrogen.

(4) The angular momentum of the electron in the first Bohr orbit of deuterium is more than that of hydrogen.

Q15

The Binding energy per nucleon of ${}_{3}^{7}Li$ and ${}_{2}^{4}He$ nucleon are 5.60 MeV and 7.06 M eV, respectively. In the nuclear reaction ${}_{3}^{7}Li + {}_{1}^{1}H \rightarrow {}_{2}^{4}He + {}_{2}^{4}He + Q$, the value of energy Q released is

(1) 19.6 MeV

(2) -2.4 MeV

(3) 8.4 MeV

(4) 17.3 MeV

Q16

A carrier signal $\mathrm{C}(\mathrm{t}) = 25\sin(2.512 \times 10^{10}\mathrm{t})$ is amplitude modulated by a message signal $\mathrm{m}(\mathrm{t}) = 5\sin(1.57 \times 10^{8}\mathrm{t})$ and transmitted through an antenna.What will be the bandwidth of the modulated signal ?

(1) 8 GHz

(2) 2.01 GHz

(3) 1987.5 MHz

(4) 50 MHz

Q17

The efficiency of the Carnot engine is 50% and the temperature of the sink is 500 K. If temperature of the source is kept constant and its efficiency is raised to 60%, then the required temperature of the sink will be

(1) 100 K

(2) 600 K

(3) 400 K

(4) 500 K

Q18

The two plates of a parallel plate capacitor are 4 mm apart. A slab of dielectric constant 3 and thickness 3 mm is introduced between the plates with its faces parallel to them. The distance between the plates is so adjusted that the capacitance of the capacitor becomes 2/3rd of its original value. What is the new distance between the plates?

(1) 9 mm

(2) 21 mm

(3) 5 mm

(4) 8 mm

Q19

Magnetic field at the centre of a circular coil of radius R due to i flowing through it is B. The magnetic field at a point along the axis at distance R from the centre is

(1) $\frac{B}{2}$

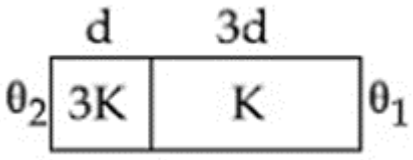
(2) $\frac{B}{4}$

(3) $\frac{B}{\sqrt{8}}$

(4) $\sqrt{8B}$

Q20

Two materials having coefficients of thermal conductivity 3K and K and thickness d and 3d respectively, are joined to form a slab as shown in the figure. The temperatures of the outer surfaces are θ_2 and θ_1 respectively, $(\theta_2 > \theta_1)$. The temperature at the interface is



(1) $\frac{\theta_2+\theta_1}{2}$

(2) $\frac{\theta_1}{6}+\frac{5\theta_2}{6}$

(3) $\frac{\theta_1}{3}+\frac{2\theta_2}{3}$

(4) $\frac{\theta_1}{10}+\frac{9\theta_2}{10}$

Q21

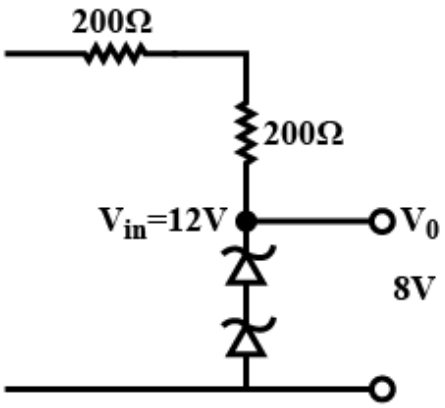
Gas at pressure P_0 is contained in a vessel. If the mass of all the molecules are halved and their speed is doubled then the resulting pressure becomes nP_0. What is the value of ' n'.

Q22

A biconvex lens made of glass $\left(\mu_g=1.5\right)$ has radius of curvature $\frac{2}{3}$m. It is kept in contact with a plane mirror. Find magnitude of power of optical device formed if space between them is filled with water $\left(\mu_w=4/3\right)$ in dioptre.

Q23

In the circuit shown below, is working as a 8 V dc regulated voltage source. When 12 V is used as an input, the power dissipated (in mW) in each diode is (Considering both zener diodes are identical)

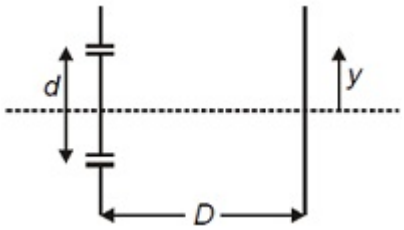


Q24

An inductor of 10 mH is connected to a 20 V battery through a resistor of 10 kΩ and a switch. After a long time, when maximum current is set up in the circuit, the current is switched off. The current in the circuit after 1 μs is $\frac{x}{100}$ mA. Then x is equal to _____ . (Take $e^{-1} = 0.37$)

Q25

In YDSE if incident light consists of two wavelengths λ_1 =4000 Å and λ_2 =5600 Å and is parallel to line SO. The minimum distance y upon screen, measured from point O, will be where the bright fringe due to two wavelengths coincide is $\frac{n\lambda_1 D}{d}$. Find n.

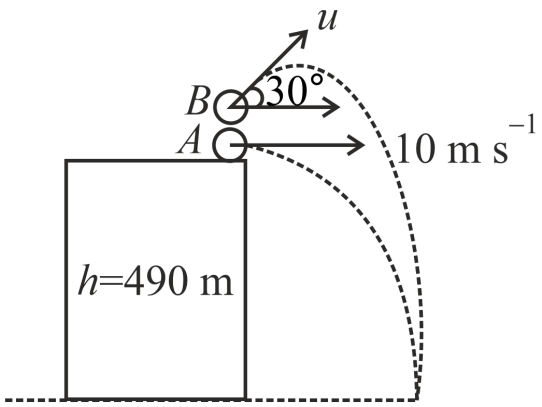


Q26

The surface of a metal is illuminated alternately with photons of energies $E_1 = 4$ eV and $E_2 = 2.5$ eV respectively. The ratio of maximum speeds of the photoelectrons emitted in the two cases is 2. The work function of the metal in (eV) is.....

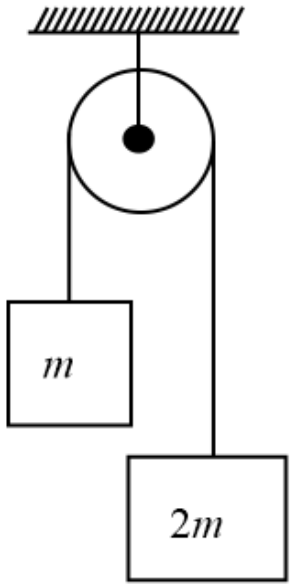
Q27

Two balls A and B are projected from the same height as shown in the figure. If the horizontal range of both the balls remains the same, then the speed u of the second ball is $\ldots\ldots$ m s^{-1} (Round off to the nearest integer)



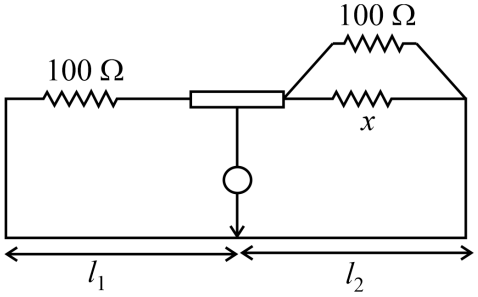
Q28

A light inextensible string that goes over a smooth fixed pulley as shown in the figure connects two blocks of masses 0.36 kg and 0.72 kg. Taking $g = 10$ m s^{-2}, find the work done (in joules) by the string on the block of mass 0.36 kg during the first second after the system is released from rest.



Q29

In a practical wheat stone bridge circuit as shown, when one more resistance of 100 Ω is connected in parallel with unknown resistance X, then ratio $\frac{l_1}{l_2}$ becomes 2. l_1 is balance length. AB is a uniform wire. Then the value of X(in Ω) must be:



Q30

A force $\overrightarrow{F} = \left(\hat{i} + 2\hat{j} + 3\hat{k}\right)$ N acts at a point $\left(4\hat{i} + 3\hat{j} - \hat{k}\right)$ m. Then the magnitude of torque about the point $\left(\hat{i} + 2\hat{j} + \hat{k}\right)$ m will be $\sqrt{x}\,\mathrm{N} - \mathrm{m}$.The value of x is.....

Q31

Which of these molecules do not obey the octet rule in any of the important lewis structures needed, to describe the bonding in the molecule?

(I) $\underline{\mathrm{N}}\mathrm{O}_2$

(II) $\underline{\mathrm{As}}\mathrm{Cl}_3$

(III) $\underline{\mathrm{S}}\mathrm{F}_4$

(IV) $\underline{\mathrm{Cl}}\mathrm{O}_3^-$

(V) $\mathrm{C}_2\mathrm{H}_4$

(1) Only III and IV

(2) Only II and V

(3) Only IV

(4) I, III, IV

Q32

Among the following, the false statement is:

(1) Latex is a colloidal solution of rubber particles which are positively charged

(2) Tyndall effect can be used to distinguish between a colloidal solution and a true solution.

(3) It is possible to cause artificial rain by throwing electrified sand carrying charge opposite to the one on clouds from an aeroplane.

(4) Lyophilic sols can be coagulated by adding an electrolyte.

Q33

Which of the following pairs of ions is isoelectronic and isostructural?

(1) CO_3^{2-}, NO_3^-

(2) ClO_3^-, CO_3^{2-}

(3) SO_3^{2-}, NO_3^-

(4) ClO_2^-, SO_3^{2-}

Q34

Match List I with List II : (Both having metallurgical terms)

	List-I		List-II
(a)	Concentration of Ag ore	(i)	Reverberatory furnace
(b)	Blast furnace	(ii)	Pig iron
(c)	Blister copper	(iii)	Leaching with dilute NaCN solution
(d)	Froth floatation method	(iv)	Sulfide ores

Choose the correct answer from the options given

below:

(1) $(a)-(iii),\ (b)-(ii),\ (c)-(i),\ (d)-(iv)$

(2) $(a)-(iii),\ (b)-(iv),\ (c)-(i),\ (d)-(ii)$

(3) $(a)-(iv),\ (b)\ -(i),\ (c)-(iii),\ (d)-(ii)$

(4) $(a)-(iv),\ (b)-(iii),\ (c)-(ii),\ (d)-(i)$

Q35

Which of the following orders is correct?

a. Order of percentage relative abundance is: protium $>$ deuterium $>$ tritium.

b. Order of density is: protium $<$ deuterium $<$ tritium.

c. Order of boiling point is: protium $>$ deuterium $>$ tritium.

d. Order of atomic mass is: protium $>$ deuterium $>$ tritium.

(1) Both a and b.

(2) a, c and d.

(3) b, c and d.

(4) a, b and c.

Q36

In the given below alkali metal ions which has lowest ionic mobility in aqueous solutions?

(1) Rb^+

(2) Cs^+

(3) Li^+

(4) Na^+

Q37

Select incorrect statement :

(1) $\left[\mathrm{Ni(en)_3}\right]^{2+}$ is less stable than $\left[\mathrm{Ni(NH_3)_6}\right]^{2+}$

(2) Increase in stability of the complexes due to presence of multidentate cyclic ligand is called macrocyclic effect

(3) A complex ion that exchanges ligands slowly is said to be non-labile or inert

(4) For a given ion and ligand, greater the charge on the metal ion, greater the stability

Q38

The correct order of ionic radius is (atomic number of $\mathrm{Ce} = 58,\ \mathrm{Pr} = 59,\ \mathrm{Eu} = 63$ and $\mathrm{Dy} = 66$)

(1) $\mathrm{Pr}^{+3} > \mathrm{Ce}^{+3} > \mathrm{Dy}^{+2} > \mathrm{Eu}^{+3}$

(2) $\mathrm{Ce}^{3+} > \mathrm{Pr}^{+3} > \mathrm{Eu}^{+3} > \mathrm{Dy}^{+3}$

(3) $\mathrm{Dy}^{+3} > \mathrm{Eu}^{+3} > \mathrm{Ce}^{+3} > \mathrm{Pr}^{+3}$

(4) $\mathrm{Eu}^{+3} > \mathrm{Ce}^{+3} > \mathrm{Dy}^{+3} > \mathrm{Pr}^{+3}$

Q39

What will be the standard cell potential of galvanic cell with the following reaction?

$2\mathrm{Cr}_{(s)} + 3\mathrm{Cd}^{2+}_{(aq)} \rightarrow 2\mathrm{Cr}^{3+}_{(aq)} + 3\mathrm{Cd}_{(s)}$

[Given $\mathrm{E^\circ}_{\mathrm{Cr^{3+}/Cr}} = -0.74\mathrm{V}$ and $\mathrm{E^\circ}_{\mathrm{Cd^{2+}/Cd}} = -0.40\mathrm{V}$]

(1) 0.74 V

(2) 1.14 V

(3) 0.34 V

(4) -0.34 V

Q40

Eutrophication causes

(1) reduction in dissolved oxygen

(2) decreases BOD

(3) decrease of nutrients in water

(4) reduction in water pollution

Q41

A chromatography column, packed with silica gel as stationary phase, was used to separate a mixture of compounds consisting of (A) benzanilide (B) aniline and (C) acetophenone. When the column is eluted with a mixture of solvents, hexane : ethyl acetate $(20:80)$, the sequence of obtained compounds is:

(1) $(\mathrm{B}), (\mathrm{C})$ and (A)

(2) $(\mathrm{B}), (\mathrm{A})$ and (C)

(3) $(\mathrm{C}), (\mathrm{A})$ and (B)

(4) $(\mathrm{A}), (\mathrm{B})$ and (C)

Q42

Consider the following properties/tests,

I. Silver-mirror test.

II. Liebermann's nitroso reaction.

III. Fehling's test.

IV. Baeyer's test.

Fructose does not give

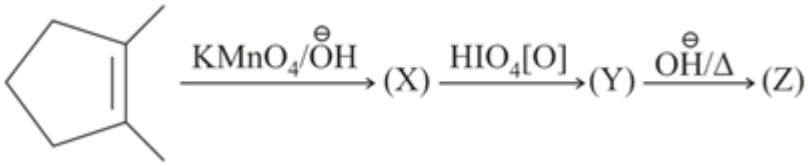
(1) II, IV

(2) I, III

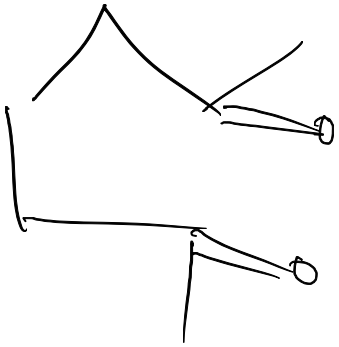
(3) I, II, III

(4) I, II, III and IV

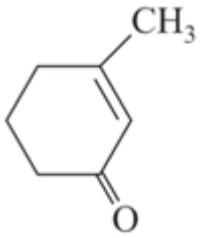
Q43



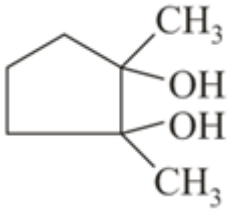
The product (Z) of the above reaction is:



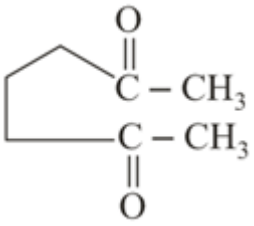
(1)



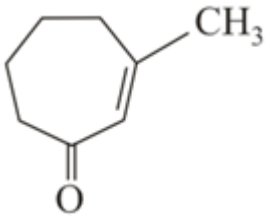
(2)



(3)

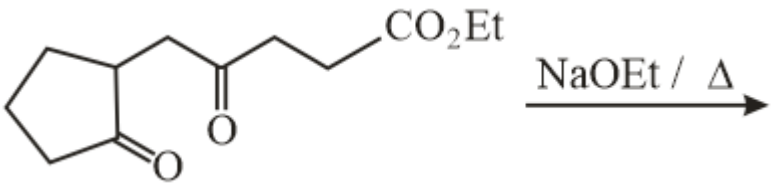


(4)

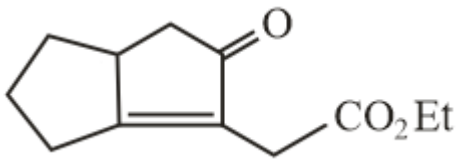


Q44

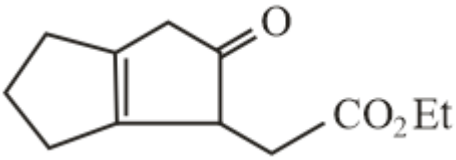
The major product obtained in the following reaction is:



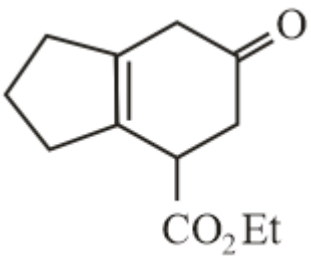
(1)



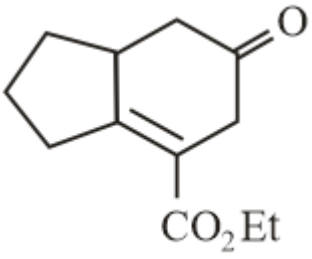
(2)



(3)



(4)



Q45

An azeotropic mixture of chloroform and diethyl ether possesses boiling point higher than either that of chloroform or diethyl ether. The mixture shows:

(1) Negative deviation from Raoult's law

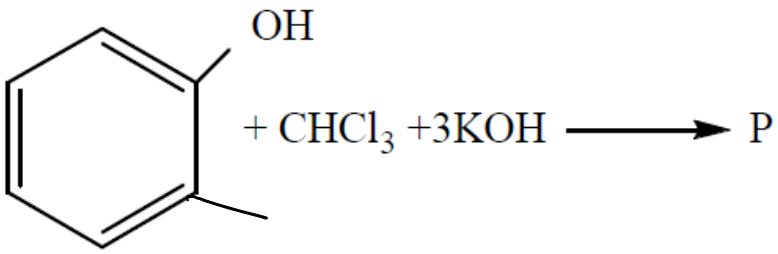
(2) Positive deviation from Raoult's law

(3) No deviation from ideal behaviour

(4) That the solution is saturated

Q46

The molecular mass difference between phenol and product P is 'X'. Find 'X'.



(1) 32

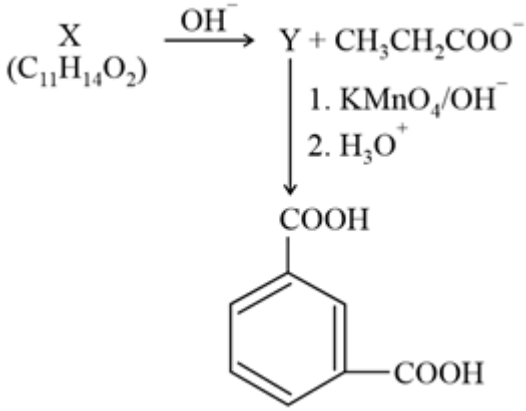
(2) 28

(3) 30

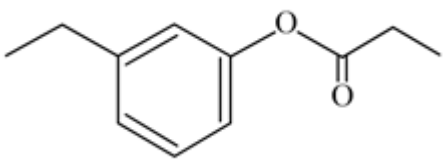
(4) 24

Q47

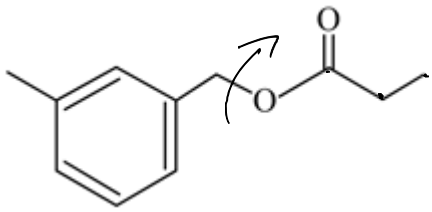
Some reactions given below ,identify the compund X .



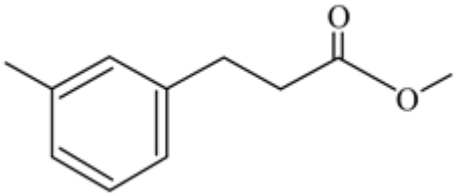
(1)



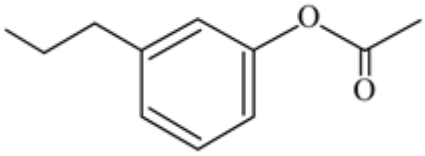
(2)



(3)



(4)



Q48

The artificial sweetener that has the highest sweetness value in comparison to cane sugar is:

(1) Sucralose

(2) Aspartame

(3) Saccharin

(4) Alitame

Q49

The reaction $C_2H_6\ (g) \rightleftharpoons C_2H_4\ (g) + H_2\ (g)$ is at equilibrium in a closed vessel at 1000 K. The enthalpy change (ΔH) for the reaction is $137.0\ \text{kJ}\ \text{mol}^{-1}$ Which one of the following actions would shift the equilibrium to the right?

(1) Decreasing the volume of the closed reaction vessel.

(2) Decreasing the temperature at which, the reaction is performed.

(3) Adding an inert gas to the closed reaction vessel.

(4) Increasing the volume of the closed reaction vessel.

Q50

Assertion: Rayon is a semisynthetic polymer whose properties are better than natural cotton.

Reason: Mechanical and aesthetic properties of cellulose can be improved by acetylation.

(1) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion.

(2) Both assertion and reason are correct, and the reason is the correct explanation for the assertion.

(3) Assertion in incorrect statement, but the reason is correct.

(4) Both assertion and reason are incorrect.

Q51

Urea is manufactured on large scale by passing CO_2 (g) through ammonia solution followed by crystallization. CO_2 for the above reaction is prepared by combustion of hydrocarbons. If combustion of 236.1 kg of a saturated hydrocarbon produces as much CO_2 as required for production of 1000 kg of urea, deduce the molecular formula of hydrocarbon. If hydrocarbon is CnH_{2n+2}. Then, find the value of n.

Q52

Consider a cube 1 of body-centered cubic unit cell of edge length 4 pm. Now atom at the body center can be viewed to be lying on the corner of another cube 2. Find the volume common to cube 1 and cube 2 in pm^3.

Q53

The quantum number 6 corresponding to the excited state of He^+ ion if on de-excitation to the ground state that ion emits photons in succession with two wavelengths only. The quantum number of the shell in which the electron comes first before occupying ground state is .....

Q54

The standard enthalpies of formation of Al_2O_3 and CaO are $-1675\ kJ\ mol^{-1}$ and $-635\ kJ\ mol^{-1}$ respectively.

For the reaction

$3CaO + 2Al \rightarrow 3Ca + Al_2O_3$ the standard reaction enthalpy $\Delta_r H^0 =$ _____ kJ.

(Round off to the Nearest Integer).

Q55

A flask has 'N' gas molecules out of which $\frac{N}{10}$ Molecules are moving at Z ms^{-1}, $\frac{N}{30}$ at Y ms^{-1} and the remaining ones are moving at X ms^{-1}. If, rms velocity of the gas is $4.89\ ms^{-1}$. Under same conditions, what is X?

Q56

What volume (in mL) of $\frac{N}{20}$ CH_3COONa should be mixed to $250\ mL$, $\frac{N}{10}$ CH_3COOH get a solution of $pH = 5$.

$K_a(CH_3COOH) = 1.8 \times 10^{-5}$

Q57

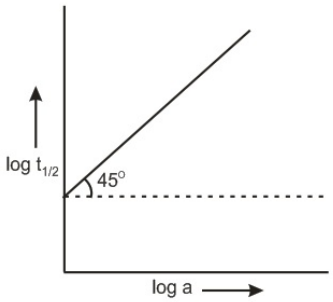
For the electrolytic production of $NaClO_4$ from $NaClO_3$ as per reaction:

$ClO_3^- + H_2O \rightarrow ClO_4^- + 2H^+ + 2e^-$

How many faradays of electricity would be required to produce 1 mole of NaClO ?

Q58

Following is the graph between log $t_{1/2}$ and log a (a = initial concentration) for a given reaction at $27^\circ C$. Hence, the order is



Q59

An isomer of the complex $Co(en)_2(H_2O)Cl_2Br$, on reaction with concentrated H_2SO_4 (dehydrating agent) it suffers loss in weight and on reaction with $AgNO_3$ solution, it gives a white precipitate which is soluble in $NH_{3(aq)}$.If one mole of complex is treated with excess $Pb(NO_3)_2$ solution, then the number of moles of white precipitate (of $PbCl_2$) formed will be X. The vale of 2X is?

Q60

The number of moles of $KMnO_4$ that will be needed to react with one mole of sulphite ion in acidic solution is x. Then the value of 10x is .....

Q61

Let W denotes the words in the English dictionary. If the relation R is given by

$R = \{(x,\ y) \in W \times W :\ \text{the word } x \text{ and } y \text{ have at least one letter in common}\}$, then R is

(1) reflexive, symmetric and not transitive

(2) reflexive, symmetric and transitive

(3) reflexive, not symmetric and transitive

(4) not reflexive, symmetric and transitive

Q62

The number of polynomials of the form x^3+ax^2+bx+c $(\forall a,b,c \in \{1\ ,2,3,.....,10\})$ which are divisible by x^2+1 is equal to

(1) 5

(2) 10

(3) 20

(4) 100

Q63

There are ten seats out of which four are to be occupied. The number of ways of arranging four persons on these four seats such that each person has exactly one neighbour is

(1) ${}^4P_2 \times {}^7P_2$

(2) ${}^4P_2 \times {}^7C_2$

(3) ${}^4C_2 \times {}^7C_2$

(4) ${}^4C_2 \times {}^7P_2$

Q64

The number obtained after dividing the number formed by the last three digits of 17^{256} by 100 is

(1) 8.61

(2) 6.81

(3) 4.31

(4) 8.31

Q65

Three numbers $a,\ b$ and c are in between 2 and 18 such that $2,\ a,\ b$ are in arithmetic progression and $b,\ c,\ 18$ are in geometric progression. If $a+b+c=25$, then the value of $c-a$ is

(1) 4

(2) 3

(3) 7

(4) 0

Q66

For $f(x)= x^2 - 2\left|x\right|$, test the continuity and differentiability of $g(x)$ in the interval [-2, 3] , where $g(x)= \begin{cases} \min\{f(t);\ -2 \leq t \leq x\} & -2 \leq x < 0 \\ \max\{f(t); 0 \leq t \leq x\} & 0 \leq x \leq 3 \end{cases}$, then

(1) $g(x)$ is continuous and differentiable everywhere except $x = 0$

(2) $g(x)$ is non - differentiable at $x = 0$ and 2

(3) $g(x)$ is non - differentiable at $x = 0,\ 1,\ 2$

(4) None of the above

Q67

If $I_1 = \int_0^1 \frac{x^{7/2}(1-x)^{9/2}}{30}dx$ and $I_2 = \int_0^1 \frac{x^{7/2}(1-x)^{9/2}}{(x+5)^{10}}dx$ and $\frac{I_1}{I_2} = 5a^3\sqrt{a}$, where $a \in N$, then the value of a is:

(1) 24

(2) 26

(3) 28

(4) 30

Q68

If $\int_0^{\pi} \frac{\sin x(\sin x+1)e^{\sin x+\cos x}}{e^{\cos x}+1}dx = a + b\int_0^{\pi} e^{\sin x}dx$ where a and b are positive rational numbers, then find the value of $100\left(a^2 + b^2\right)$.

(1) 105

(2) 120

(3) 125

(4) 75

Q69

The area (in sq. units) bounded by the curve $f(x)=\begin{cases} e^{-|x|} & |x|\geq 1 \\ |x|-1+\frac{1}{e} & |x|<1 \end{cases}$ and the x-axis is

(1) $2+\frac{2}{e^2}$

(2) $2+\frac{1}{e^2}$

(3) $\frac{1}{e^2}$

(4) $1+\frac{2}{e^2}$

Q70

The solution of the differential equation $\frac{dy}{dx}+\frac{xy}{1-x^2}=x\sqrt{y}$, $(|x|<1)$ is $\sqrt{y}=-\frac{f(x)}{3}+C\left(1-x^2\right)^{\frac{1}{4}}$, where $f\left(\frac{1}{2}\right)=\frac{3}{4}$ and C is an arbitrary constant. Then, the value of $f\left(-\frac{1}{2}\right)$ is

(1) $-\frac{3}{4}$

(2) $\frac{3}{4}$

(3) $\frac{1}{4}$

(4) $\frac{3}{2}$

Q71

If $f(x)$ is a differentiable function satisfying $|f'(x)|\leq 4\ \forall x\in[0,4]$ and $f(0)=0$, then

(1) $f(x)=18$ has no solution in $x\in[0,4]$

(2) $f(x)=18$ has more than 2 solutions in $x\in[0,4]$

(3) $f(x)=14$ has no solution in $x\in[0,4]$

(4) $f(x)=20$ has 2 solution in $x\in[0,4]$

Q72

The coordinates of the orthocenter of the triangle that has the coordinates of midpoints of its sides as

$(0,0),\ (1,2)$ and $(-6\ ,3)$ is

(1) $(0,0)$

(2) $(-4,5)$

(3) $(-5,5)$

(4) $(-4,4)$

Q73

From a point P, two tangents PA and PB are drawn to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If these tangents cut the coordinate axes at 4 concyclic points, then the locus of P is

(1) $x^2-y^2=\left|a^2-b^2\right|$

(2) $x^2-y^2=a^2+b^2$

(3) $x^2+y^2=\left|a^2-b^2\right|$

(4) $x^2+y^2=a^2+b^2$

Q74

Let $\vec{a}$ and $\vec{b}$ are two non-collinear vector such that $\left|\vec{a}\right|=1$. The angles of a triangle whose two sides are represented by vectors $\sqrt{3}\left(\vec{a}\times\vec{b}\right)$ and $\vec{b}-\left(\vec{a}\cdot\vec{b}\right)\vec{a}$ are

(1) $\frac{\pi}{2},\frac{\pi}{4},\frac{\pi}{4}$

(2) $\frac{\pi}{2},\frac{\pi}{3},\frac{\pi}{6}$

(3) $\frac{\pi}{2},\frac{5\pi}{12},\frac{\pi}{12}$

(4) $\frac{\pi}{4},\frac{\pi}{3},\frac{5\pi}{12}$

Q75

The number of integral points on the circle, touching the parabola $y^2=8x$ at $(2,4)$ and passing through $(0,4)$, are equal to

(1) 1

(2) 2

(3) 4

(4) 3

Q76

The image of the line $\frac{x}{2}=\frac{y-1}{5}=\frac{z+1}{3}$ in the plane $x+y+2z=3$ meets the $xz-$plane at the point (a,b,c), then the value of c is equal to

(1) $\frac{11}{6}$

(2) $\frac{129}{6}$

(3) $\frac{115}{6}$

(4) $\frac{232}{3}$

Q77

Let $A=\{1,2,3,4,5,6,7\}$. The number of surjective functions defined from A to A such that $f(i)=i$ for atleast four values of i from $i=1,2,\ldots\ldots,7$, is:

(1) 7 !

(2) 92

(3) 126

(4) 407

Q78

The mean square deviation of a set of observations $\mathrm{x}_1, \mathrm{x}_2 \ldots \ldots \mathrm{x}_{\mathrm{n}}$ about a point m is defined as $\frac{1}{\mathrm{n}} \sum_{\mathrm{i}=1}^{\mathrm{n}}\left(\mathrm{x}_{\mathrm{i}}-\mathrm{m}\right)^2$. If the mean square deviations about -1 and 1 of a set of observations are 7 and 3 respectively. The standard deviation of those observations is

(1) $\sqrt{2}$

(2) 2

(3) 5

(4) $\sqrt{3}$

Q79

The negation of $p \wedge(q \rightarrow \sim r)$ is

(1) $\sim p \wedge(q \wedge r)$

(2) $p \vee(q \vee r)$

(3) $p \vee(q \wedge r)$

(4) $\sim p \vee(q \wedge r)$

Q80

If the value of $(1+\tan 1^{\circ})(1+\tan 2^{\circ})(1+\tan 3^{\circ}) \ldots \ldots \ldots (1+\tan 44^{\circ})(1+\tan 45^{\circ})$ is 2^{λ}, then the sum of the digits of the number λ is

(1) 3

(2) 6

(3) 5

(4) 4

Q81

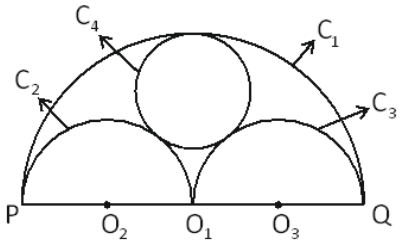
The equation of the plane passing through the point of intersection of the lines $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-3}{2}$, $\frac{x-3}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ and perpendicular to the line $\frac{x-2}{2}=\frac{y-3}{3}=\frac{z-2}{1}$ is $P=0$. If the distance of the point $(1,1,3)$ from $P=0$ is k units, then the value of $\frac{k^2}{2}$ is equal to

Q82

If $\vec{a_1}, \vec{a_2}, \vec{a_3}$ and $\vec{b_1}, \vec{b_2}, \vec{b_3}$ be two sets of non-coplanar vectors, such that $\vec{a_p} \cdot \vec{b_q}=\begin{cases}0 & \text{if } p \neq q \\ 4 & \text{if } p=q\end{cases}$ for $p=1,2,3$ and $q=1,2,3$, then the value of $\left[\vec{a_1} \quad 2\vec{a_2} \quad 3\vec{a_3}\right]\left[\vec{b_1}+\vec{b_2} \quad \vec{b_2}+\vec{b_3} \quad \vec{b_3}+\vec{b_1}\right]$ is equal to

Q83

In the figure $PQ,\ PO_1$ and O_1Q are the diameters of semicircles $C_1,\ C_2$ and C_3 with centres at $O_1,\ O_2$ and O_3 respectively and the circle C_4 touches the semicircles C_1, C_2 and C_3. If $PQ=24$ units and the area of the circle C_4 is A sq. units, then the value of $\frac{16\pi}{A}$ is equal to (here, $PO_1=O_1Q$)



Q84

The value of $\lim\limits_{x \to \frac{\pi}{2}} \left(1^{\frac{1}{\cos^2 x}} + 2^{\frac{1}{\cos^2 x}} + \ldots\ldots\ldots + 10^{\frac{1}{\cos^2 x}}\right)^{\cos^2 x}$ is equal to

Q85

Let a_1, a_2, a_3,....., a_{11} be real numbers satisfying $a_1 = 15$, $27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}\ \forall\ k = 3, 4,$ .....,11. If $\frac{a_1^2 + a_2^2 + \ldots. + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \ldots. + a_{11}}{11}$ is equal to

Q86

Consider the system of equations $ax + y + bz = 0,\ bx + y + az = 0$ and $ax + by + abz = 0$ where $a, b \in \{0,1,2,3,4\}$. The number of ordered pairs (a, b) for which the system has non-trivial solutions is

Q87

Let matrix $\mathrm{A} = \begin{bmatrix} \mathrm{x} & \mathrm{y} & -\mathrm{z} \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$, where $\mathrm{x},\ \mathrm{y},\ \mathrm{z} \in \mathrm{N}$. If $|\mathrm{adj}|\mathrm{adj}|\mathrm{adj}|\mathrm{adjA}||||= 4^8 \cdot 5^{16}$, then the number of such matrices A is equal to (where, $|M|$ represents determinant of a matrix M)

Q88

Let the points A, B, C and D are represented by complex numbers Z_1, Z_2, Z_3 and Z_4 respectively. If A, B and C are not collinear and $2Z_1 + Z_2 + Z_3 - 4Z_4 = 0$, then the value of $\frac{\text{Area of } \Delta DBC}{\text{Area of } \Delta ABC}$ is equal to K. Find the value of 4K.

Q89

If p is a positive integer and f be a function defined for positive numbers and attains only positive values such that $f(xf(y)) = x^p y^4$ then find p.

Q90

The tangent at any point P of a curve c meets the x axis at Q whose abscissa is positive and OP=OQ, O being the origin. Curve c passes through the point (1, 0) and tangent on point (1, 0) is given by x=a. Then a equals to.....

Answer Key

Q1 (3)	**Q2** (3)	**Q3** (4)	**Q4** (4)
Q5 (2)	**Q6** (1)	**Q7** (3)	**Q8** (1)
Q9 (3)	**Q10** (4)	**Q11** (4)	**Q12** (4)
Q13 (2)	**Q14** (1)	**Q15** (4)	**Q16** (4)
Q17 (3)	**Q18** (4)	**Q19** (3)	**Q20** (4)
Q21 (2)	**Q22** (2)	**Q23** (40)	**Q24** (74)
Q25 (7)	**Q26** (2)	**Q27** (11)	**Q28** (8)
Q29 (100)	**Q30** (195)	**Q31** (4)	**Q32** (1)
Q33 (1)	**Q34** (1)	**Q35** (1)	**Q36** (3)
Q37 (1)	**Q38** (2)	**Q39** (3)	**Q40** (1)
Q41 (3)	**Q42** (1)	**Q43** (1)	**Q44** (1)
Q45 (1)	**Q46** (2)	**Q47** (2)	**Q48** (4)
Q49 (4)	**Q50** (2)	**Q51** (12)	**Q52** (8)
Q53 (2)	**Q54** (230)	**Q55** (4)	**Q56** (900)
Q57 (2)	**Q58** (0)	**Q59** (1)	**Q60** (4)
Q61 (1)	**Q62** (2)	**Q63** (1)	**Q64** (2)

Q65 (3)	**Q66** (2)	**Q67** (4)	**Q68** (3)
Q69 (4)	**Q70** (2)	**Q71** (1)	**Q72** (3)
Q73 (2)	**Q74** (2)	**Q75** (3)	**Q76** (2)
Q77 (2)	**Q78** (4)	**Q79** (4)	**Q80** (3)
Q81 (7)	**Q82** (768)	**Q83** (1)	**Q84** (10)
Q85 (0)	**Q86** (6)	**Q87** (36)	**Q88** (2)
Q89 (2)	**Q90** (1)		

Q1

As, $\beta = 2.1 \times 10^9\ N/m^2$

Also given that $\frac{\Delta V}{V} = 0.1\% = 10^{-4}$

We know that, $\beta = \frac{pV}{\Delta V}$

$\Rightarrow \quad p = \beta \left(\frac{\Delta V}{V}\right)$

$= 2.1 \times 10^9 \times 10^{-4}$

$= 2.1 \times 10^5\ N/m^2$

Q2

Obviously, the magnetic moment of an atom is due to both orbital motion and spin motion. Thus the assertion statement is a correct statement.

While all the charges do not produce a magnetic field, only moving charges produce a magnetic field. Thus the reason statement is incorrect.

Q3

Here, E stands for emf of the cells and r is the internal resistance of the cell.

Case I:

$E + E = (r + r + 5) \times 1.0$

$\Rightarrow 2E = 2r + 5$ ...(i)

Case II:

$E = \left(\frac{r \times r}{r + r} + 5\right) \times 0.8$

$\Rightarrow E = \left(\frac{r}{2} + 5\right) 0.8$

$\Rightarrow E = 0\ .4\ r + 4\ .0$ ...(ii)

Multiplying equation (ii) by two and equating with equation (i), we get,

$2r + 5 = 0.8r + 8$

$\Rightarrow 1.2r = 3$

$\Rightarrow r = \frac{3}{1.2} = 2.5\ \Omega$

Q4

(1) Pressure, Young's modulus and stress

Pressure and stress are defined as the force acting per unit area, $[\mathrm{P}]=[\sigma]=\frac{\mathrm{MLT}^{-2}}{\mathrm{L}^2}=\mathrm{ML}^{-1}\,\mathrm{T}^{-2}$.

Young's modulus is defined as $Y=\frac{\text{Linear stress}}{\text{Linear strain}}$. Since linear strain is dimensionless, thus Young's modulus has the same dimension of stress.

(2) E.M.F., potential difference and electric potential are energy difference.

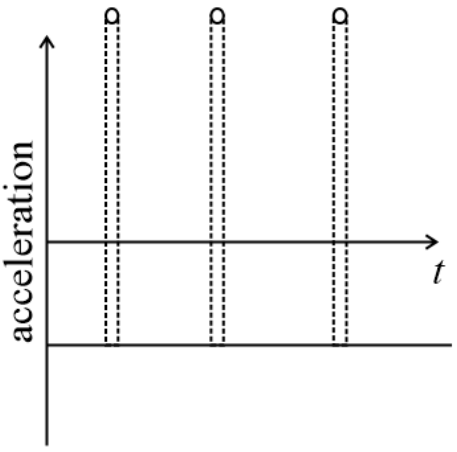
(3) Heat, work done and energy are all forms of energy thus, has the same dimensions $\mathrm{ML}^2\,\mathrm{T}^{-2}$.

(4) Dipole moment, electric flux and electric field have different units.

Q5

Gravitational force always operates on the body in the downward direction during the free fall of the ball, and thus the ball's acceleration lies in the downward vertical direction, which is constant over time. At the collision of the ball with the surface, the acceleration turns positive at a single span of time. The ground imposes an upward impulse and thus the upward acceleration on the ball that makes the ball move upward, and yet it accelerates afterward in the downward direction as fast as the ball loses contact with the ground.

Thus, the correct plot of ball is



Q6

Acceleration of body$= g\sin\theta - \mu g\cos\theta$

$= 9.8[\sin 45^\circ - 0.5\cos 45^\circ] = \frac{4.9}{\sqrt{2}}\ \text{m sec}^{-2}$

Q7

From conservation of angular momentum.

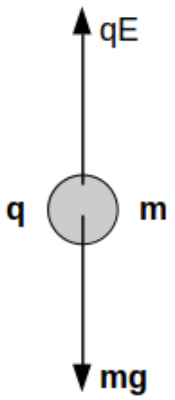
$\mathrm{mv_0R_0} = \mathrm{mv'}\left(\frac{\mathrm{R_0}}{2}\right)$

$\Rightarrow \mathrm{v'} = 2\mathrm{v_0}$

Hence, final $\mathrm{KE} = \frac{1}{2}\mathrm{mv'^2} = \frac{1}{2}\mathrm{m}(2\mathrm{v_0})^2$

$= 2\mathrm{mv_0^2}$

Q8



For equilibrium of charged oil drop,

$qE = mg$

$\therefore\ q = \frac{mg}{E} = \frac{\left(9.9\times10^{-15}\right)\times10}{\left(3\times10^{4}\right)} = 3.3\times10^{-18}\ \text{C}.$

Q9

From Work Energy Theorem,

Change in Kinetic energy (KE)= Work done

$KE_f - KE_i =$ area under the graph of F vs x

$KE_f - 0 = 5$

$KE_f = 5\ \text{J}$

Q10

Since there is no external torque acting on the planet revolving around sun, so when a planet revolves around the sun, its angular momentum $\left(L = I\omega = mr^2\omega\right)$ remains conserved. Here, I is the moment of inertia,ω is the angular velocity, m is the mass of the planet and r is the orbit radius.

Angular momentum of planet when it is at minimum distance $L_1 = mr_{min}^2\omega$

Let the angular velocity of the planet at maximum distance be ω_2. Then angular momentum at maximum distance $L_2 = mr_{max}^2\omega_2$.

Using $L_1 = L_2$,

$\Rightarrow mr_{min}^2\omega = mr_{max}^2\omega_2$

$\Rightarrow \omega_2 = \left(\frac{r_{\min}}{r_{\max}}\right)^2\omega$

Q11

Instantaneous current in AC circuit, at instant t

(Assuming $I = 0$ at $t = 0$).

$I = I_0\ \sin(wt) = I_0\ \sin(2\pi ft)$

$I_0,\ w\ and\ f$ are peak current, angular frequency and frequency, respectively.

We know, rms current $I_{rms} = \frac{I_0}{\sqrt{2}}$. if rms current is equal to the instantaneous current at time t, then

$I = \frac{I_0}{\sqrt{2}} = I_0\sin(wt),\ \sin(wt) = \frac{1}{\sqrt{2}}$

$\Rightarrow wt = 2\pi ft = \frac{\pi}{4}$

$\Rightarrow t = \frac{\pi}{4\left(2\pi f\right)} = \frac{\pi}{4\left(2\pi\times 50\right)} = \frac{1}{400}$

$\Rightarrow t = 2.5\ \text{ms}$

Q12

Let the equation of two waves are

$y_1 = a\sin(\omega t - kx) \qquad \ldots(i)$

And $y_2 = a\sin(\omega t - kx + \phi)\ \ldots(ii)$

When they superpose, the resultant wave is

$y = y_1 + y_2$

$= a[\sin(\omega t - kx) + \sin(\omega t - kx + \phi)]$

$= a\left[2\sin\left(\omega t - kx + \frac{\phi}{2}\right)\cos\left(-\frac{\phi}{2}\right)\right]$

$= 2a\sin\left(\omega t - kx + \frac{\phi}{2}\right)\cos\frac{\phi}{2}$

$= \left(2a\cos\frac{\phi}{2}\right)\sin\left(\omega t - kx + \frac{\phi}{2}\right) \ \ldots(iii)$

Comparing Eq. (iii) with (i) or (ii), we get

$a = 2a\cos\frac{\phi}{2} \Longrightarrow \cos\frac{\phi}{2} = \frac{1}{2}$

$\Longrightarrow \ \cos\frac{\phi}{2} = \cos\frac{\pi}{3}$

$\therefore \ \cos\frac{\phi}{2} = \cos\frac{\pi}{3}$

$\therefore \ \frac{\phi}{2} = \frac{\pi}{3}$

or $\phi = \frac{2\pi}{3}$

Q13

Given that,

$B_0 = 10^{-4}$ T,

the speed of the electromagnetic wave is

$c = \frac{E_0}{B_0}$

$(c = 3 \times 10^8 \text{ m s}^{-1})$

$E_0 = cB_0$

$E_0 = 3 \times 10^8 \times 10^{-4}$

$E_0 = 3 \times 10^4 \text{ V m}^{-1}$

Q14

We know that, radius of Bohr orbit is $r_{\mathrm{n}}=\left(\frac{n^2h^2}{4\pi^2mkZe^2}\right)$

where, m is the reduced mass of the electron.

$m = \frac{m_eM}{m_e+M}$

where, m_e is the mass of the electron and M is the mass of the nucleus.

The mass of deuterium is more than that of hydrogen. So, the reduced mass of electron is more for deuterium than that for hydrogen.

Hence, the radius of first Bohr orbit of deuterium is less than that of hydrogen.

Q15

${}_4Li^7 + {}_1H^1 \rightarrow {}_2\left({}_2He^4\right)$

BE of products$= ((5.6\text{ MeV})\times 7) + 0$

$= 39.2\text{ MeV}$

$E_i = -39.2\text{ MeV}$

BE of reactant$=(7.06)\times 4 \times 2$

$= 56.48\text{ MeV}$

$E_f = -56.48\text{ MeV}$

As nuclear energy decreases, some energy will be released.

$Q_{\text{release}} = E_i - E_f =(-39.2)-(-56.48)= 17.28\text{ MeV}$

Q16

Band width $= 2\mathrm{f_m}$

$\omega_{\mathrm{m}} = 1.57 \times 10^8 = 2\pi \mathrm{f_m}$

$\mathrm{BW} = 2\mathrm{f_m} = \frac{10^8}{2}\text{ Hz} = 50\text{ MHz}$

Q17

Efficiency (η) of a Carnot engine is given by $\eta = 1 - \frac{T_2}{T_1}$, where T_1 is the temperature of the source and T_2 is the temperature of the sink.

Here, $T_2 = 500$ K.

$\therefore \ 0.5 = 1 - \frac{500}{T_1} \Rightarrow T_1 = 1000 \ \text{K}$

Now, $\eta' = 0.6 = 1 - \frac{T_2'}{1000}$ (T_2' is the new sink temperature)

$T_2' = 400$ K

Q18

Here, distance between parallel plates d = 4 mm = 0.004 m, K = 3, thickness t = 3 mm = 0.003 m and d_1 =?

$\therefore C = \frac{\varepsilon_0 A}{d}$ and $C_1 = \frac{\varepsilon_0 A}{d_1 - t\left(1 - \frac{1}{K}\right)}$

since $C_1 = \frac{2}{3}C$ (given)

$$\therefore \ \frac{\varepsilon_0 A}{d_1 - t\left(1 - \frac{1}{K}\right)} = \frac{2}{3}\frac{\varepsilon_0 A}{d}$$

$$\frac{1}{d_1 - t\left(1 - \frac{1}{K}\right)} = \frac{2}{3d}$$

$$\frac{1}{d_1 - 0.003\left(1 - \frac{1}{3}\right)} = \frac{2}{3 \times 0.004}$$

$$\frac{1}{d_1 - 0.003 \times \frac{2}{3}} = \frac{1}{0.006}$$

$$\frac{1}{d_1 - 0.002} = \frac{1}{0.006}$$

$$d_1 - 0.002 = 0.006$$

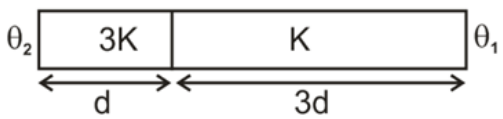
$$d_1 = 0.006 + 0.002 = 0.008 \text{ m} = 8 \text{ mm}.$$

Q19

The magnetic field at a point along the axis at distance R from the centre of a circular coil of radius R carrying i is, $B = \frac{\mu_0 i R^2}{2\left(R^2 + R^2\right)^{3/2}}$, by using formula magnetic field at axial point at a distance equal to the radius of coil.

$$= \frac{\mu_0 i}{2\sqrt{8}R} = \frac{B}{\sqrt{8}} \left[B_{\text{centre}} = B = \frac{\mu_0 i}{2R}\right]$$

Q20



Let the temperature of the junction T $^\circ$C .

Rate of heat flow in Rod 1 = rate of heat flow in Rod 2

$\frac{3\,\mathrm{kA}}{\mathrm{d}}(\theta_2-\mathrm{T})=\frac{\mathrm{kA}}{3\mathrm{d}}(\mathrm{T}-\theta_1)$

$\Rightarrow 9(\theta_2-\mathrm{T})=(\mathrm{T}-\theta_1)$

$\Rightarrow 10\mathrm{T}=9\theta_2+\theta_1$

$\Rightarrow \mathrm{T}=\frac{9\theta_2+\theta_1}{10}=\frac{\theta_1}{10}+\frac{9\theta_2}{10}$

Q21

Pressure of a gas is given by $P=\frac{1}{3}\frac{mN}{V}(v_{\mathrm{rms}})^2$.

Where, m =mass of the gas,

$N=$ Number of gas molecules,

V =Volume of the vessel,

$v_{\mathrm{rms}}=$ RMS speed of gas molecules.

So, $P_0=\frac{1}{3}\frac{mN}{V}(v_{\mathrm{rms}})^2$.

If the mass of all the molecules are halved and their speed is doubled,

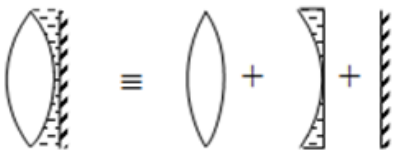
$P=\frac{1}{3}\frac{(m/2)N}{V}(2v_{\mathrm{rms}})^2$

$\Rightarrow\ P=2\left[\frac{1}{3}\frac{mN}{V}(v_{\mathrm{rms}})^2\right]$

$\Rightarrow\ P=2P_0$

Therefore, $n=2$.

Q22



focal length of convex mirror $(f_1)=Rm$

focal length of concave lens $(f_2)=-3Rm$

effective lens $(f_{eq})=\frac{f_1f_2}{f_1+f_2}=\frac{-3R}{-2}=\frac{3R}{2}=1\mathrm{m}$

$\frac{1}{f}=\frac{1}{f_M}-\frac{2}{f_l}=\frac{1}{\infty}-\frac{2}{1}$

$f=-\frac{1}{2}\mathrm{m}$

$P=-2D$

Q23

$i=\frac{(12-8)}{(200+200)}\mathrm{A}=\frac{4}{400}=10^{-2}\ \mathrm{A}$

Power loss in each diode$=(4)\left(10^{-2}\right)\ \mathrm{W}=40\ \mathrm{mW}$

Q24

$I_{\max}=\frac{V}{R}=\frac{20\ \mathrm{V}}{10\ \mathrm{K}\Omega}=2\ \mathrm{mA}$

For LR-decay circuit,

$I=I_{\max}e^{-Rt/L}$

$I=2\ \mathrm{mA}\ \mathrm{e}^{\frac{-10\times10^{3}\times1\times10^{-6}}{10\times10^{-3}}}$

$I=2\ \mathrm{mA}\ \mathrm{e}^{-1}$

$I=2\times0.37\ \mathrm{mA}$

$I=\frac{74}{100}\ \mathrm{mA}$

$x=74$

Q25

$n_1\overline{X}_1=n_2\overline{X}_2$

$n_1\frac{D\lambda_1}{d}=n_2\frac{D\lambda_2}{d}$

$n_1\lambda_1=n_1\lambda_2$

$\frac{\lambda_1}{\lambda_2}=\frac{n_2}{n_1}=\frac{4000}{5600}=\frac{40}{58}=\frac{5}{7}$

$\therefore\quad \frac{n_2}{n_1}=\frac{5}{7}\Rightarrow\frac{X_1}{X_2}=\frac{7}{5}$

$\therefore\quad y=7\overline{X}_1=\frac{7D\lambda_1}{d}$

Q26

Given,

Energy of two photons are $E_1=4\ \mathrm{eV}$ and $E_2=2.5\ \mathrm{eV}$

The ratio of maximum speeds of the photoelectrons emitted in the two cases is $\frac{v_1}{v_2} = 2$

Using Einstein equation of photoelectric effect,

$KE_{max} = \frac{1}{2}mv^2 = E - \phi \ \ldots \left(1\right)$

Where, ϕ is the work function of metal and E is the energy of photon

Now using equation for both the cases we get,

$\frac{1}{2}mv_1^2 = 4 - \phi \ \ \ldots \left(2\right)$

$\frac{1}{2}mv_2^2 = 2.5 - \phi \ \ldots \left(3\right)$

Dividing equation (2) and (3) and substitute given values, we get,

$\frac{{v_1}^2}{{v_2}^2} = \frac{4-\phi}{2.5-\phi} = (2)^2$

$\Rightarrow 3\phi = 6$

$\Rightarrow \phi = 2 \ \text{eV}$

Q27

The ball, B, follows horizontal and angular projectile and the ball A follows only horizontal projectile,

the height of the tower is, $h = 490$ m, and both the particle follows the same range,

now for particle A,

$R = u\sqrt{\frac{2h}{g}} = 10 \times \sqrt{\frac{2\times 490}{9.8}} = 100 \text{ m}$

and for oblique projectile,

$R = 100 \text{ m} = u\text{cos}\theta \times t + u\text{cos}\theta \times \left(490 + \frac{u\sin^2\theta}{2g}\right)$

it means,

$R = u\text{cos}\theta \times \frac{u\sin\theta}{g} + u\text{cos}\theta \times \sqrt{\frac{\left(490 + \frac{u^2\sin^2\theta}{2g}\right)}{g}}$

$\Rightarrow R = 100 = u\cos 30^\circ \times \frac{u\sin 30^\circ}{9.8} + u\cos 30^\circ \times \sqrt{\frac{\left(490 + \frac{u^2\sin^2 30^\circ}{2\times 9.8}\right)}{9.8}}$

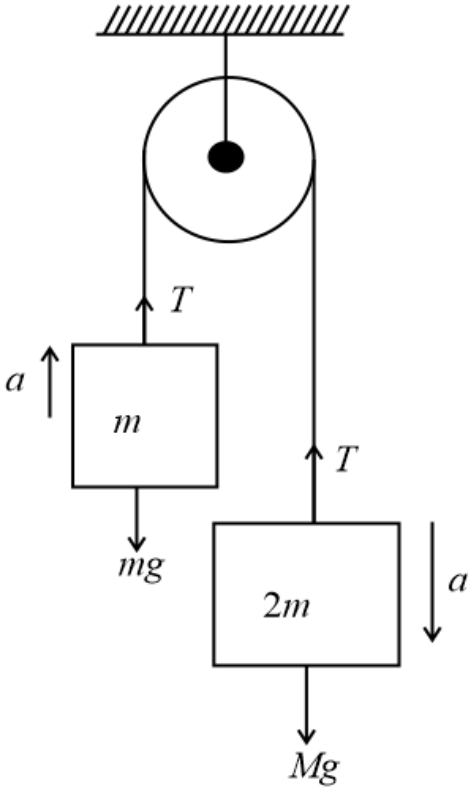
$\Rightarrow u = 10.9 \text{ m s}^{-1}$

Q28

Given, $m = 0.36$ kg and $M = 2m = 0.72$ kg

Let a be the acceleration when the system is released.

Forces on m and M are shown in figure.



From the figure, we have

$T - mg = ma \quad \ldots(1)$

and $Mg - T = Ma \quad \ldots(2)$

Adding above two equation,

$g(M - m) = (M + m)a$

$\Rightarrow a = \frac{g(M-m)}{(M+m)}$

$a = \frac{g(0.72-0.36)}{(0.72+0.36)} = \frac{g \times 0.36}{1.08} = \frac{g}{3}$

Putting these values in (1),

$T = \left(m \times \frac{g}{3}\right) + (m \times g)$

$T = \frac{4mg}{3}$

Now, displacement of block is $s = ut + \frac{1}{2}at^2$

Here, initial velocity $u = 0$, then $s = \frac{1}{2}at^2$.

Work done by the string on the block is $W = T \times s = T \times \frac{1}{2}at^2$

$= \frac{4mg}{3} \times \frac{1}{2} \times \frac{g}{3} \times t^2$

$= \frac{4 \times 0.36 \times 10}{3} \times \frac{1}{2} \times \frac{10}{3} \times 1$

$W = 8$ J

Q29

The given Wheat stone's bridge is in a balanced condition

$\frac{P}{Q} = \frac{R}{S}$

$\Rightarrow \frac{100}{l_1} = \frac{\left(\frac{100\,x}{100+x}\right)}{l_2}$

$\because \quad \frac{l_1}{l_2} = 2$ So, $\frac{100}{\left(\frac{100x}{100+x}\right)} = 2$

$\Rightarrow$ The unknown resistance is $\mathrm{x} = 100\ \Omega$

Q30

$\overrightarrow{r} = (4-1)\hat{i} + (3-2)\hat{j} + (-1-1)\hat{k}$

$= 3\hat{i} + \hat{j} - 2\hat{k}$

$\tau = \overrightarrow{r} \times \overrightarrow{F} = \begin{vmatrix} \hat{i} & j & \hat{k} \\ 3 & 1 & -2 \\ 1 & 2 & 3 \end{vmatrix}$

$= \hat{i}(7) - \hat{j}(11) + \hat{k}(5) = 7\hat{i} - 11\hat{j} + 5\hat{k}$

$= \sqrt{49 + 121 + 25} = \sqrt{195}$

Q31

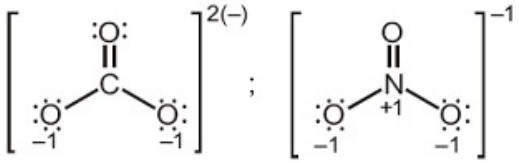
Octet rule: Central atom should have eight electrons in its valence shell in the bonded state.

- Around N in NO_2 has 7 electrons in valence shell
- Around S in SF_4 has 10 electrons in valence shell
- Around Cl in ClO_3^- has 12 electrons in the valence shell

Q32

Latex is a colloidal solution of rubber particles which are negatively charged not positively charged.

Q33



Molecules contain the same number of electrons are known as isoelectronic species and those have same shape known as isostructural. Above two compounds, Both have 32 electrons with trigonal planar structure.

Q34

(a) Concentration of Ag is performed by leaching with dilute NaCN solution

(b) Pig iron is formed in blast furnace

(c) Blister Cu is produced in Bessemer converter

(d) Froth floatation method is used for sulphide

ores.

Note : During extraction of Cu reverberatory furnace is involved.

Q35

Relative abundance:

It is the existence of a naturally occurring element in a percentage of atoms with a particular atomic weight or molar mass.

There are three isotopes of hydrogen:

Protium (H), deuterium (D) and tritium (T).

Order of abundance: $\mathrm{H} > \mathrm{D} > \mathrm{T}$

Order of density: $\mathrm{H} < \mathrm{D} < \mathrm{T}$

Order of boiling point: $\mathrm{H} < \mathrm{D} < \mathrm{T}$

Order of atomic mass: $\mathrm{H} < \mathrm{D} < \mathrm{T}$

Q36

As we know that hydration power decreases on moving down the group hence among alkali metals Li has excessive hydration & hence it has low mobility in aqueous solution.

Q37

$\left[\mathrm{Ni(en)_3}\right]^{2+}$ contains chelating ligand whereas $\left[\mathrm{Ni(NH_3)_6}\right]^{2+}$ contains ammonia molecule as ligand which can't form chelate. Due to this former is more stable than the latter.

Q38

Due to increase in effective nuclear charge on moving left to right in periodic table, size decreases.

$\mathrm{Ce}^{+3} > \mathrm{Pr}^{+3} > \mathrm{Eu}^{+3} > \mathrm{Dy}^{+3}$

Q39

The standard cell potential can be calculated by using this formula,

$\mathrm{E}^{\mathrm{o}}_{\mathrm{cell}} = \mathrm{E}^{\mathrm{o}}_{\mathrm{cathode}} - \mathrm{E}^{\mathrm{o}}_{\mathrm{anode}}$

Given, $\mathrm{E}^{\circ}{}_{\mathrm{Cr}^{3+}/\mathrm{Cr}} = -0.74\ \mathrm{V}$ and $\mathrm{E}^{\circ}{}_{\mathrm{Cd}^{2+}/\mathrm{Cd}} = -0.40\ \mathrm{V}$

$\mathrm{E}^{\mathrm{o}}_{\mathrm{cell}} = \mathrm{E}^{\mathrm{o}}_{\mathrm{Cd}^{2+}/\mathrm{Cd}} - \mathrm{E}^{\mathrm{o}}_{\mathrm{Cr}^{3+}/\mathrm{Cr}}$

$= -0.40 - (-0.74) = +0.34\ \mathrm{V}$

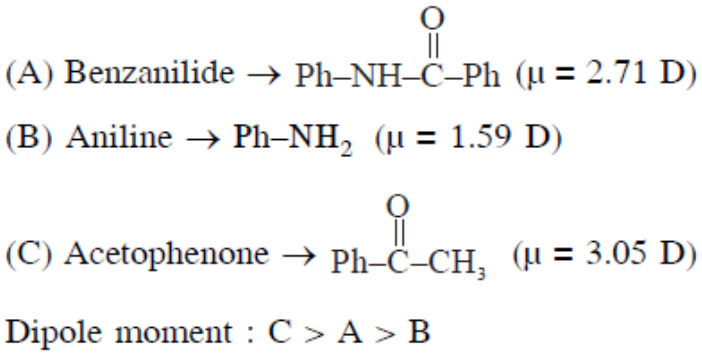
Q40

Eutrophication refers to excessive increases in minerals and nutrients in aquatic system, resulting in excess of algae growth and reduction in dissolved oxygen.

Q41

A dipole moment arises in any system in which there is a separation of charge. They can, therefore, arise in ionic bonds as well as in covalent bonds. Dipole moments occur due to the difference in electronegativity between two chemically bonded atoms.

A bond dipole moment is a measure of the polarity of a chemical bond between two atoms in a molecule. It involves the concept of electric dipole moment, which is a measure of the separation of negative and positive charges in a system.



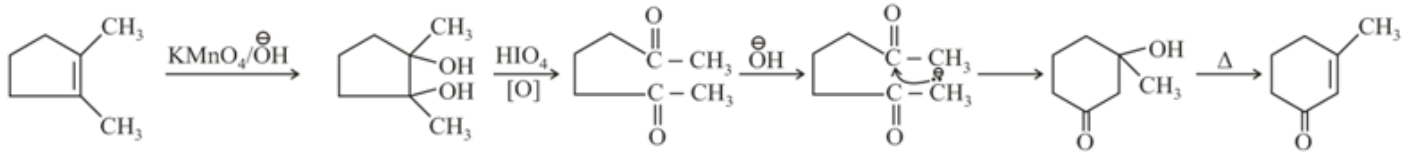
Hence the sequence of obtained compounds is (C), (A) and (B)

Q42

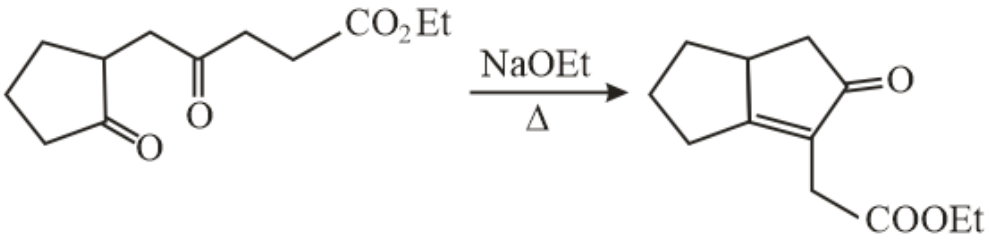
- Baeyer's reagent, named after the German organic chemist Adolf von Baeyer, is used in organic chemistry as a qualitative test for the presence of unsaturation, such as double bonds.
- While phenol is reacted with $NaNO_2$ and concentrated H_2SO_4, it provides a deep green or blue colour which changes to red on dilution with water. While generated alkaline along with NaOH original green or blue colour is restored. This reaction is termed as Liebermann's nitroso reaction and is employed as a test of phenol.
- Fructose is a reducing sugar, it can reduce both Tollen's reagent and Fehling's solution

Q43

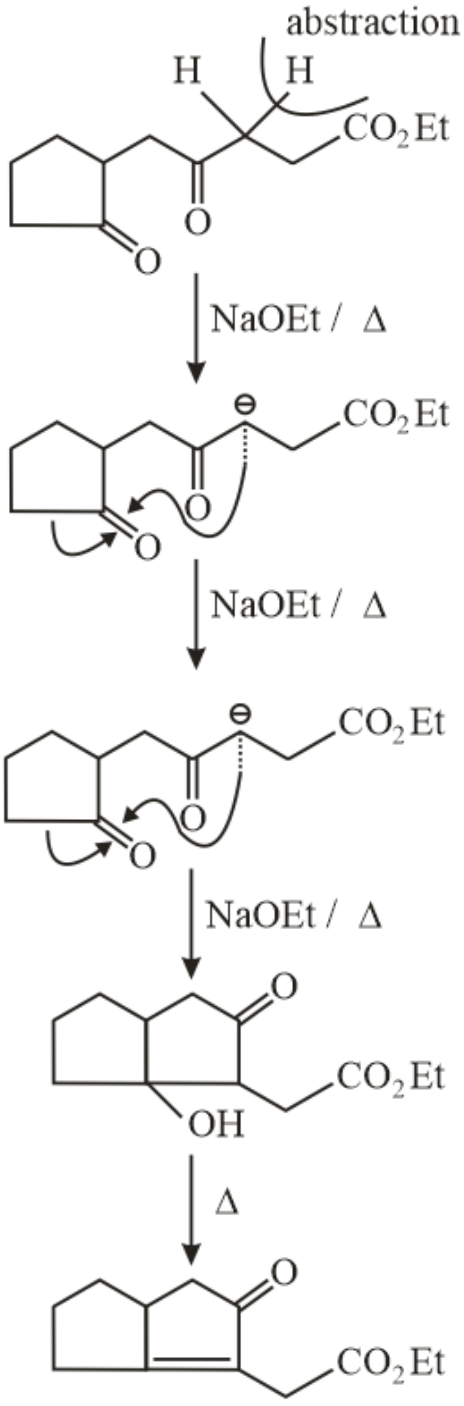
Initially potassium permanganate forms diol which are oxdised by periodic acid and forms diketone. This diketone later shows intramolecular aldol condensation reaction with hydroxide ion and later removes water to form alkene in presence of heat as shown below:



Q44

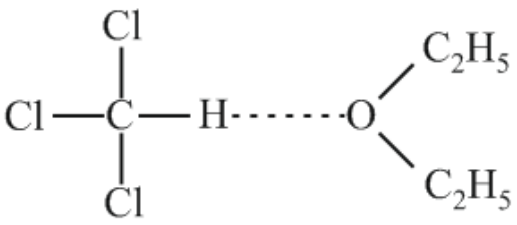


Intramolecular aldol condensation



Q45

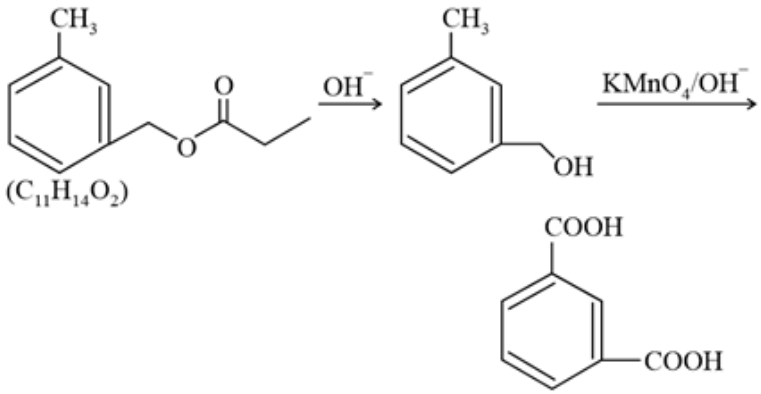
Azeotropic mixtures with higher boiling point show negative deviation. There occurs H-bonding between chloroform and diethyl ether in the solution, hence the escaping tendency of either component becomes less.



Q46

Reaction is Reimer Tiemann .Overall CHO will get substituted in place ortho position and resultant compound is called salicylaldehyde. Difference in molecular mass between phenol and salicylaldehyde is due to carbon and oxygen which is equal to 12+16=28

Q47



1) First reaction is hydrolysis of ester in basic medium. Ester on hydrolysis gives Carboxylic acid and Alcohol.

2) Second reaction is side chain Oxidation of Benzene in presence of $KMnO_4$ to give Carboxylic acid group.

Q48

Alitame is an artificial sweetener that has 1000 times the sweetness value of cane sugar.

Q49

$C_2H_6\ (g) \rightleftharpoons C_2H_4\ (g) + H_2\ (g);\ \ \Delta H = 137.0\ kJ/mol$

$\Delta ng = (1+1) - 2 = -1$ (Negative)

As ΔH is positive, increasing temperature will shift reaction forward (right) direction and as Δ ng is negative, increasing the volume of container or decreasing pressure can shift reaction forward.

Q50

The strength of cellulose is improved by acetylation and then used for making packing material. Rayon (semisynthetic polymer of cellulose) has superior properties than natural cotton.

Q51

Urea manufacture

$CO_2 + 2\,NH_3 \longrightarrow H_2NCONH_2 + H_2O$

44 g 60 g

For 1000 Kg urea CO_2 required is

$$\frac{44}{60} \times 1000 \times 10^3 = 733.33\text{g} \times 10^3$$

$$C_nH_{2n+2^+} \quad \left(n + \frac{2n+2}{4}\right) O_2 \rightarrow n\,CO_2 + (n+1)\,H_2O\ (\ell)$$

$$n = \frac{733.33}{44} \times 10^3 \text{ moles}$$

Apply POAC to C

$$n\left(\frac{236.1\times10^3}{12n+2n+2}\right) = \frac{733.33}{44} \times 10^3 \times 1$$

$$\frac{n}{14n+2} \times 236.1 = 16.67$$

$236.1\,n = 233.38\,n + 33.34$

$2.72\,n = 33.34$

$$n = \frac{33.34}{2.72} \cong 12$$

Q52

Given,

Edge length (a) for BCC unit cell for cube $1 = 4$ pm



Body-centered cubic structure

In BCC, number of atoms per unit cell $= 2$

(atoms occupy corner of the cube as well as the body centre of cube).

Volume of cube $=\mathrm{a}^3$

Volume for cube 1 when edge length $(\mathrm{a}) = 4\ \mathrm{pm}$

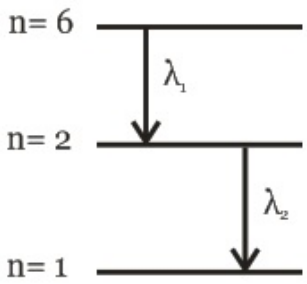
Volume $= (\mathrm{a})^3 = (4)^3 = 64\ \mathrm{pm}^3$

Now atom at the body centre of cube 1 can be viewed to be lying on the corner of another cube 2,

Side length of common volume $=\frac{\mathrm{a}}{2} = \frac{4}{2} = 2\,\mathrm{pm}$

Now, the volume common to cube 1 and cube 2 $= \left(\frac{\mathrm{a}}{2}\right)^3 = \left(2\right)^3 = 8\ \mathrm{pm}^3$

Q53



It first comes to energy level no. 2 before de-excitation.

Q54

Given reaction:

$3\mathrm{CaO} + \mathrm{Al} \rightarrow \mathrm{Al_2\,O_3} + 3\,\mathrm{Ca}$

Now, $\Delta_\mathrm{r}\mathrm{H}^\circ = \Sigma\Delta_\mathrm{f}\mathrm{H}^\circ_{\text{Products}} - \Sigma\Delta_\mathrm{f}\mathrm{H}^\circ_{\text{Reactants}}$

$= [1 \times (-1675) + 3 \times 0] - [3 \times (-635) + 2 \times 0]$

$= +230\ \mathrm{kJ\ \ mol^{-1}}$

Q55

$$U_{rms} = \sqrt{\frac{3RT}{M}}$$

$$U_{mp} = \sqrt{\frac{2RT}{M}}$$

X represents most probable velocity.

If we divide rms value with mp speed.

We get the ratio as shown below:

$$U_{rms} = 1.2248\ U_{mp}$$

$$\frac{4.89}{1.2248} = 4 ms^{-1}$$

Q56

For acetic acid and sodium acetate, Normality = Molarity (as n−factor is 1)

$$n_{CH_3COONa} = \frac{M}{10} \times V = {}^{V}/_{20} \text{ mmoles}$$

$$n_{CH_3COOH} = \frac{M}{10} \times 250 = 25 \text{ mmoles}$$

Now, the total volume $= (250 + V)$ mL

Concentration of $CH_3COONa = \frac{{}^{V}/_{20}}{(250+V)}$

Concentration of $CH_3COOH = \frac{25}{(250+V)}$

Here in the question, an acidic buffer solution is forming. Now, for the equilibrium, $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

$$1.8 \times 10^{-5} = \frac{\frac{V}{20} \times 10^{-5}}{25}$$

$$\Rightarrow 1.8 = \frac{V}{500}$$

$$\Rightarrow V = 1.8 \times 500 = 900 \text{ mL}$$

Q57

$$ClO_3^- + H_2O \rightarrow ClO_4^- + 2H^+ + 2e^-$$

To produce one mole of ClO_4^-, 2 mole of electron are required.

Therefore, number of Faraday for production of 1 mole of $NaClO_4 = 2$ F

Q58

$\mathrm{A} \rightarrow$ Products

From graph $\rightarrow$

$\log t_{\frac{1}{2}} \propto \tan 45 (= \log a)$

$\log t_{\frac{1}{2}} \propto \log a$

$t_{\frac{1}{2}} \propto a$ (Initial Condition)

For zero-order reaction

$\therefore t_{\frac{1}{2}} \propto (a)^{1-n}$

Q59

In any complex compounds, only the group present outside the coordination sphere as counter ions show their tests. Thus,

1. Since the compound is showing loss in weight with dehydrating agent, it must have water molecule outside the coordination sphere.
2. Since the complex is giving white precipitate with $AgNO_3$, it must have atleast one Cl^- ion outside the coordination sphere.
3. Co has a coordination number six, Thus the total sum of denticities of all ligands inside the coordination sphere should be six.

The formula that fits into all above criterias is-

$[CoBrCl(en)_2]Cl.H_2O$

Clearly, one mole of compound will give one mole chloride ion on dissolving in water.

Now, two moles of chlroide ions on reaction with one mole of $Pb(NO_3)_2$ gives one mole of $PbCl_2$.

$2Cl^- + Pb(NO_3)_2 \rightarrow 2NO_3^- + PbCl_2 \downarrow$

Thus, with one mole of chloride ion, we will get only 0.5 moles of lead(II) chloride.

Therefore, X = 0.5

i.e. 2X = 1

Q60

In acidic medium, every molecule of $KMnO_4$ accepts 5 - electron and SO_2 is oxidized to SO_4^{2-} by releasing two electron per SO_2 molecular.

In acidic medium, $KMnO_4$ oxidises sulphite into sulphate

$$\left[\overset{(+4)2-}{SO_3} \rightarrow \overset{(+6)}{SO}_4^{2-} + 2e^-\right] \times 5$$

$$\left[MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O\right] \times 2$$

5 moles of $SO_3^{2-} \equiv 2$ moles of $KMnO_4$

1 mole of $SO_3^{2-} \equiv \frac{2}{5}$moles of $KMnO_4$

Q61

Given, $R = \{(x, y) \in W \times W$: the word x and y have atleast one letter in common$\}$

Let, $W = \{\text{cat},\ \text{toy},\ \text{you}, \ldots\}$

Clearly, R is reflexive and symmetric but not transitive.

[Since, cat R toy, toy Ryou $\nRightarrow$ cat R you]

Q62

$f(x) = x^3 + ax^2 + bx + c$ is divisible by $x^2 + 1$ or $(x + i)(x - i)$

So, $f(i) = 0$ and $f(-i) = 0$

$f(i) = 0 \Rightarrow i^3 + ai^2 + bi + c = 0$

$\Rightarrow -i - a + bi + c = 0$

$\Rightarrow (c - a) + i(b - 1) = 0 \Rightarrow c = a, b = 1$

$f(-i) = 0 \Rightarrow (-i)^3 + a(-i)^2 + b(-i) + c = 0$

$\Rightarrow i - a - ib + c = 0$

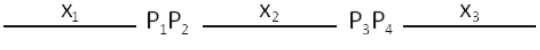
$\Rightarrow (c - a) - i(b - 1) = 0 \Rightarrow c = a,\ b = 1$

$b = 1$ and $c = a$

$\Rightarrow$ 10 polynomials are possible

Q63

Group of two persons can be selected in ${}^4C_2$ ways. Let P_1P_2 are together and P_3P_4 are together. Let x_1 be the number of seats vacant at the left end, x_2 seats are between two pairs and x_3 seats are vacant at the right end.



We have, $x_1 + x_2 + x_3 = 6$, where $x_1,\ x_3 \geq 0$ and $x_2 \geq 1$

$\Rightarrow x_1+(x_2+1)+x_3=6,$ where $x_1,\ x_2,\ x_3 \geq 0$

$\Rightarrow x_1+x_2+x_3=5$

Number of ways of selecting seats $= {}^{n+r-1}C_{r-1} = {}^{3+5-1}C_{3-1} = {}^{7}C_2$

Persons can interchange their seats in 2! $\times$ 2! ways

$\Rightarrow$ Required ways $= {}^{4}C_2 \times {}^{7}C_2 \times 2! \times 2! = {}^{4}P_2 \times {}^{7}P_2$

Q64

Last three digits of 17^{256}

$= (289)^{128} = (-1+290)^{128}$

$= {}^{128}C_0(-1)^{128} - {}^{128}C_1 \cdot 290 + {}^{128}C_2 \cdot (290)^2 + \ldots\ldots$

$= 1 - 128 \times 290 + 64 \times 127 \times 29^2 \times 100 + \ldots\ldots$

$= 1 - \ldots\ldots 120 + \ldots\ldots 800 + \ldots\ldots$

$= 681$

Hence, $\frac{681}{100} = 6.81$

Q65

$a+b+c=25;\ 2a=2+b;\ c^2=18b$

$\Rightarrow 2a+2b+2c=50$

$\Rightarrow 2+b+2b+2c=50$

$\Rightarrow 3b+2c=48 \Rightarrow \frac{c^2}{6}+2c=48$

$\Rightarrow c^2+12c-48\times 6=0 \Rightarrow c^2+12c-24\times 12=0$

$\Rightarrow (c+24)(c-12)=0$

$\Rightarrow c=12, -24 \Rightarrow c=12$ (between 2 and 18)

$\Rightarrow b=\frac{c^2}{18}=\frac{144}{18}=8$

$\Rightarrow a=\frac{b+2}{2}=5$

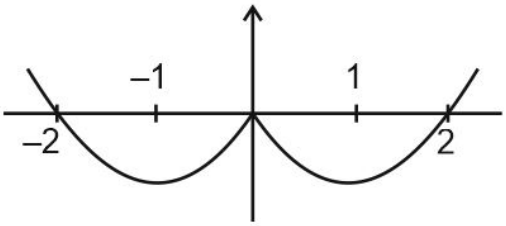
$\Rightarrow a=5,\ b=8$ and $c=12$

$\Rightarrow c-a=7$

Q66

Here $f(x)=\begin{cases} x^2+2x, & x<0 \\ 0\ , & x=0 \\ x^2-2x, & x>0 \end{cases}$

graph of f(x)



Case I :, $-2\ \le x<\ -1,\ f(x)$ **decreases** $\Rightarrow\quad g(x)=x^2+2x$

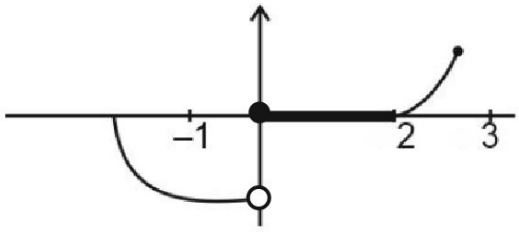
Case II : $-1\ \le x<0,\ f(x)$ **increases** $\Rightarrow\ \ g(x)=\ -1$

Case III : $0\ \le x<1,\ \ f(x)$ **decreases** $\Rightarrow g(x)=0$

Case IV : $1\ \le x<2,\ \ f(x)$ **increases but** $f(x)<0\Rightarrow\quad g(x)=f(0)=0$

Case V : $2\ \le x\ \le 3,\ \ f(x)$ **increases and** $f(x)\ge 0\ \forall\ x\ \in[2,\ 3)\Rightarrow\ \ g(x)=f(x)=x^2-2x,\ 2\ \le x\ \le 3$

graph of g(x)



Therefore $g(x)=\begin{cases} x^2+2x, & -2\ \le x<-1 \\ -1\ , & -1\ \le x<0 \\ 0\ , & 0\ \le x<2 \\ x^2-2x\ , & 2\ \le x\ \le 3 \end{cases}$

Clearly $g(x)$ **is continuous everywhere except at** $x=0$ **.**

Also $g(x)$ **is non differentiable at** $x=0$ **and 2.**

Q67

Rewrite the integral as

$I_2=\int_0^1\left(\frac{x}{5+x}\right)^{7/2}\left(\frac{1-x}{5+x}\right)^{9/2}\frac{dx}{(5+x)^2}$

and do the substitution $\frac{x}{5+x}=t$, so that $\frac{dx}{(5+x)^2}=\frac{dt}{5}$ and the integral becomes $\frac{1}{(5)^{11/2}}\int_0^{1/6}(t)^{7/2}(1-6t)^{9/2}dt$ and now from here do the substitution $6t=u$ and we simply obtain $I_2=\frac{1}{5^{9/2}\times 6^{7/2}}I_1$ and we conclude $a=30$.

Q68

Using King

$I = \int_0^\pi \frac{\sin x(1+\sin x)e^{\sin x - \cos x}}{e^{-\cos x}+1}dx$

$I = \int_0^\pi \frac{\sin x(1+\sin x)e^{\sin x}}{e^{\cos x}+1}dx$ ...(i)

Add $2I = \int_0^\pi \frac{\sin x(1+\sin x)e^{\sin x}(e^{\cos x}+1)}{e^{\cos x}+1}dx$

$I = \frac{1}{2}\int_0^\pi \sin x(1+\sin x)e^{\sin x}dx$

$I = \frac{1}{2}\int_0^\pi e^{\sin x}\left(\sin x + 1 - \cos^2 x\right)dx$

$I = \frac{1}{2}\int_0^\pi e^{\sin x}dx + \frac{1}{2}\int_0^\pi e^{\sin x}\left(\sin x - \cos^2 x\right)dx$

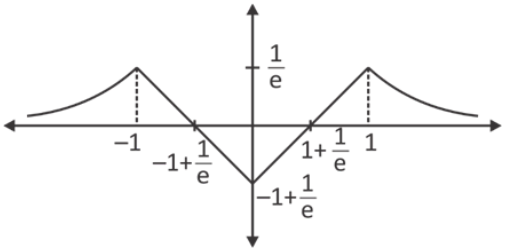
$I = \frac{1}{2}\int_0^\pi e^{\sin x}dx + \frac{1}{2}\left[e^{\sin x}(-\cos x)\right]_0^\pi$

$I = \frac{1}{2}\int_0^\pi e^{\sin x}dx + \frac{1}{2}[1+1]$

$I = 1 + \frac{1}{2}\int_0^\pi e^{\sin x}dx$

$\therefore \quad 100\left(1+\frac{1}{4}\right) = 125$

Q69



$\because$ Graph of $f(x)$ is symmetric about x-axis, hence,

Area $= 2\int_0^\infty |f(x)|dx$

$= 2\left[\frac{1}{2}\times\left(1-\frac{1}{e}\right)\left(1-\frac{1}{e}\right)+\frac{1}{2}\times\frac{1}{e}\times\frac{1}{e}+\int_1^\infty e^{-x}dx\right]$

$= 2\left[\frac{1}{2}\left[1-\frac{2}{e}+\frac{2}{e^2}\right]+[-e^{-x}]_1^\infty\right]$

$=\left[1-\frac{2}{e}+\frac{2}{e^2}+\frac{2}{e}\right]= 1+\frac{2}{e^2}$ sq. units

Q70

Given equation is $\frac{1}{\sqrt{y}}\frac{dy}{dx} + \frac{x}{1-x^2}\sqrt{y} = x$

Let, $2\sqrt{y} = \nu$

$\Rightarrow \frac{1}{\sqrt{y}}\frac{dy}{dx} = \frac{d\nu}{dx}$

Thus, we have

$\frac{d\nu}{dx} + \frac{x}{2(1-x^2)}\nu = x$

$\therefore$ I.F. $= e^{\int \frac{x}{2\left(1-x^2\right)}dx}$

$= e^{\int \frac{-d\left(1-x^2\right)}{4\left(1-x^2\right)}}$

$= e^{-\frac{1}{4}\ln\left(1-x^2\right)}$

$= \left(1-x^2\right)^{-\frac{1}{4}}$

Thus, the solution is $\nu\left(1-x^2\right)^{-\frac{1}{4}} = \int x\left(1-x^2\right)^{-\frac{1}{4}}dx$

or $\nu \cdot \left(1-x^2\right)^{-\frac{1}{4}} = -\frac{2}{3}\left(1-x^2\right)^{\frac{3}{4}} + C'$

or $2\sqrt{y} = -\frac{2}{3}\left(1-x^2\right) + C'\left(1-x^2\right)^{\frac{1}{4}}$

$\Rightarrow \sqrt{y} = -\frac{\left(1-x^2\right)}{3} + C\left(1-x^2\right)^{\frac{1}{4}}$

$\Rightarrow f(x) = 1-x^2$

$\Rightarrow f\left(-\frac{1}{2}\right) = 1-\frac{1}{4} = \frac{3}{4}$

Q71

Applying LMVT in $x \in [0,t]$ for $f(x)$, we get

$f'(c) = \frac{f(t)-f(0)}{t-0}$

$\Rightarrow |f'(c)| = \left|\frac{f(t)}{t}\right| \leq 4$

$\Rightarrow |f(t)| \leq 4t$

As $t \in [0,4]$

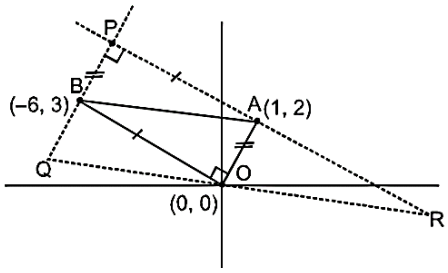
$\therefore |f(t)| \leq 16$

Q72

Line joining the midpoints of two sides is $\|$ to the third side and half of it.

$\angle \mathrm{O} = 90^{\circ}$ (as $AO \perp OB$)

Hence P will be the orthocentre

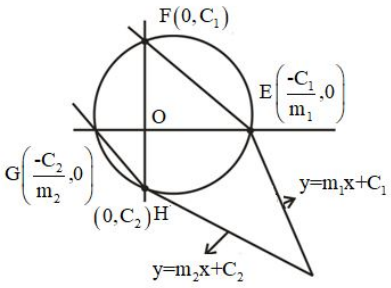


$AOBP$ forms a rectangle.

$\therefore \quad P = A + B - O$ (using the concept that diagonals bisect each other)

$P(x,y) = \begin{bmatrix} x = -6+1-0 = -5 \\ y = 2+3-0 = 5 \end{bmatrix}$

Q73



Let the equation of tangents are

$y = m_1x + \mathrm{C}_1$, and

$y = m_2x + \mathrm{C}_2$

which cuts the coordinate axes at $\mathrm{E}, \mathrm{F}, \mathrm{G}, \mathrm{H}$ as shown in the figure

Now, $\mathrm{OE} \times \mathrm{OG} = \mathrm{OF} \times \mathrm{OH}$

$\Rightarrow \left(\frac{-\mathrm{C}_1}{m_1}\right) \times \left(\frac{+\mathrm{C}_2}{m_2}\right) = (\mathrm{C}_1)(-\mathrm{C}_2) \Rightarrow m_1m_2 = 1$

Let P be (h, k) and equation of the tangent through P on the hyperbola is

$y = mx \pm \sqrt{a^2m^2 - b^2}$

$\Rightarrow \left(k^2 - mh\right)^2 = \left(a^2m^2 - b^2\right)$

$\Rightarrow \left(h^2 - a^2\right)m^2 - 2khm + k^2 + b^2 = 0$

whose roots are m_1 and $m_2 \Rightarrow m_1m_2 = \frac{k^2+b^2}{h^2-a^2} = 1$

$\Rightarrow$ locus is $x^2 - y^2 = a^2 + b^2$

Q74

Let, $\vec{A} = \sqrt{3}\left(\vec{a} \times \vec{b}\right)$

$\vec{B} = \vec{b} - \left(\vec{a} \cdot \vec{b}\right)\vec{a}$

$\vec{A} \cdot \vec{B} = 0 \Rightarrow \vec{A}$ is perpendicular to $\vec{B}$

$\left|\vec{A}\right| = \sqrt{3}\left|\vec{a} \times \vec{b}\right| = \sqrt{3}\left|\vec{a}\right|\left|\vec{b}\right|\sin\theta = \sqrt{3}\left|\vec{b}\right|\sin\theta$

$\left|\vec{B}\right|^2 = \left|\vec{b} - \left(\vec{a} \cdot \vec{b}\right)\vec{a}\right|^2$

$= \left|\vec{b}\right|^2 + \left(\vec{a} \cdot \vec{b}\right)^2\left|\vec{a}\right|^2 - 2\left(\vec{a} \cdot \vec{b}\right)^2$

$= \left|\vec{b}\right|^2 + \left|\vec{b}\right|^2\cos^2\theta - 2\left|\vec{b}\right|^2\cos^2\theta$

$= \left|\vec{b}\right|^2\sin^2\theta$

$\Rightarrow \left|\overrightarrow{B}\right| = \left|\overrightarrow{b}\right| \sin\theta$

$\frac{\left|\overrightarrow{A}\right|}{\left|\overrightarrow{B}\right|} = \frac{\sqrt{3}}{1} \Rightarrow$ other two angles are $\frac{\pi}{6}, \frac{\pi}{3}$

Q75

Equation of tangent at $(2,4)$ on the parabola $y^2 = 8x$ is

$y(4) = 8\left(\frac{x+2}{2}\right) \Rightarrow y = x + 2$

Let the equation of the circle touching line $y = x + 2$ at $(2,4)$ is

$(x-2)^2 + (y-4)^2 + \lambda(x - y + 2) = 0$ which passes through $(0, 4)$

$\Rightarrow 4 + 0 + \lambda(0 - 4 + 2) \Rightarrow \lambda = 2$

$\Rightarrow$ Required circle is $x^2 + y^2 - 2x - 10y + 24 = 0$

$\Rightarrow (x-1)^2 + (y-5)^2 = 2$

If x and y are integers, then

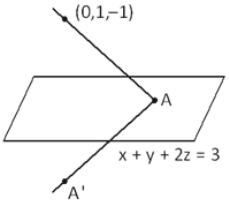
$(x-1)^2 = 1 = (y-5)^2$

$\Rightarrow x = 0, 2$ and $y = 4, 6$

$\Rightarrow 4$ integral points lie on the circle

Q76

Any general point on the line is $(2\lambda, 5\lambda + 1, 3\lambda - 1)$



On satisfying this point on the plane, we get,

$2\lambda + 5\lambda + 1 + 6\lambda - 2 = 3$

$13\lambda = 4 \Rightarrow \lambda = \frac{4}{13}$

So, coordinates of the point are $\left(\frac{8}{13}, \frac{33}{13}, \frac{-1}{13}\right)$

This point also lies on the image of the line

Image of point $(0,1,-1)$ also lies on the image of the line

$\frac{x-0}{1} = \frac{y-1}{1} = \frac{z+1}{2} = -2\frac{(-4)}{6}$

$x = \frac{4}{3}, y = \frac{7}{3}, z = \frac{5}{3}$

Point is $\left(\frac{4}{3}, \frac{7}{3}, \frac{5}{3}\right)$

Equation of image of the line is $\frac{x-\frac{4}{3}}{28} = \frac{y-\frac{7}{3}}{-8} = \frac{z-\frac{5}{3}}{68}$

For $xz-$plane, putting $y = 0$, we get,

$\frac{x-\frac{4}{3}}{28} = \frac{7}{24} = \frac{z-\frac{5}{3}}{68}$

$\Rightarrow z = \frac{129}{6}$

Q77

$$A = \{1, 2, 3, 4, 5, 6, 7\}$$

Case-I : When exactly 4 values follows $f(i) = i$

$$^7C_4 \times 3! \left(\frac{1}{2!} - \frac{1}{3!}\right) = 70$$

Case-II : When exactly 5 values follows $f(i) = i$

$$^7C_5 \times 1 = 21$$

Case-III : When all 7 values follows $f(i) = i$ number of function $= 1$

Total functions $= 70 + 21 + 1 = 92$

Q78

$\frac{\sum_{i=1}^{n} (x_i+1)^2}{n} = 7$(given)

$\Rightarrow \sum x_i^2 + 2\sum x_i + n = 7n$

$\Rightarrow \sum x_i^2 + 2\sum x_i = 6n \ldots (1)$

Also, $\frac{\sum (x_i-1)^2}{n} = 3$ (given)

$\Rightarrow \sum x_i^2 - 2\sum x_i + n = 3n$

$\Rightarrow \sum x_i^2 - 2\sum x_i = 2n \ldots (2)$

From (1) and (2)

$\sum x_i^2 = 4n, \ \sum x_i = n$

Standard deviation$= \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$

$= \sqrt{\frac{4n}{n} - 1} = \sqrt{3}$

Q79

$\sim (p \wedge (q \rightarrow \sim r)) = \sim p \vee (\sim (q \rightarrow \sim r))$

$(\because \sim(p \wedge q) = \sim p \vee \sim q \ \& \ \sim(p \rightarrow q) = p \wedge \sim q)$

$= \sim p \vee (q \wedge \sim (\sim r))$

$= \sim p \vee (q \wedge r)$

Q80

If, $A + B = 45^o$

$\tan(A + B) = 1$

$\because \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$\Rightarrow \tan A + \tan B = 1 - \tan A \tan B$

$\Rightarrow (1 + \tan A)(1 + \tan B) = 2$

LHS

$= [(1 + \tan 1^o)\ (1 + \tan 44^o)]\ [(1 + \tan 2^o)(1 + \tan 43^o)] \ldots [(1 + \tan 45^o)] \left[\text{for each} (1 + \tan\theta)\left[1 + \tan\left(\frac{\pi}{4} - \theta\right) = 2\right]\right]$

$= 2^{22}\ (1 + 1)$

$= 2^{23}$

$= 2^{\lambda}$

then, $\lambda = 23$.

Hence the sum of digits of λ is $2 + 3$

$= 5$

Q81

Any general point on the line $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ is $(3\lambda + 1, \lambda + 2, 2\lambda + 3)$

For the point of intersection, this point must satisfy the line $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

$\Rightarrow \frac{3\lambda-2}{1} = \frac{\lambda+1}{2} = \frac{2\lambda+1}{3} \Rightarrow \lambda = 1$

So, point of intersection is $(4,3,5)$

The required equation of plane is

$2(x-4)+3(y-3)+1(z-5)=0$

$\Rightarrow 2x+3y+z-22=0$

Hence, the distance of the plane from $(1,1,3)=\left|\frac{2+3+3-22}{\sqrt{14}}\right|=\sqrt{14}=k$

Q82

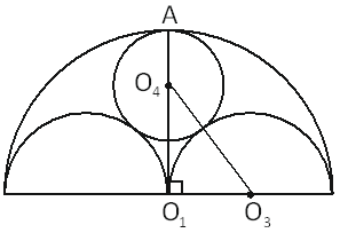
$$\left[\vec{a_1} \quad 2\vec{a_2} \quad 3\vec{a_3}\right]\left[\vec{b_1}+\vec{b_2} \quad \vec{b_2}+\vec{b_3} \quad \vec{b_3}+\vec{b_1}\right]$$

$$=6\left[\vec{a_1} \quad \vec{a_2} \quad \vec{a_3}\right]2\left[\vec{b_1} \quad \vec{b_2} \quad \vec{b_3}\right]$$

$$=12\begin{vmatrix}\vec{a_1}\cdot\vec{b_1} & \vec{a_1}\cdot\vec{b_2} & \vec{a_1}\cdot\vec{b_3}\\ \vec{a_2}\cdot\vec{b_1} & \vec{a_2}\cdot\vec{b_2} & \vec{a_2}\cdot\vec{b_3}\\ \vec{a_3}\cdot\vec{b_1} & \vec{a_3}\cdot\vec{b_2} & \vec{a_3}\cdot\vec{b_3}\end{vmatrix}$$

$$=12\begin{vmatrix}4 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 4\end{vmatrix}=12\times 64=768$$

Q83



Let the point of contact of C_4 & C_1 is A, center of C_4 is O_4 & radius is equal to r

$\Rightarrow AO_1=12\Rightarrow O_1O_4=12-r$

Also, $O_4O_3=r+6$ and $O_1O_3=6$

$\Rightarrow (r+6)^2=(12-r)^2+36$

$\Rightarrow 36r=144\Rightarrow r=4\Rightarrow A=16\pi$

$\Rightarrow \frac{16\pi}{A}=\frac{16\pi}{16\pi}=1$

Q84

Let, $L=\lim\limits_{x\to\frac{\pi}{2}}\left(1^{\frac{1}{\cos^2 x}}+2^{\frac{1}{\cos^2 x}}+\ldots\ldots\ldots+10^{\frac{1}{\cos^2 x}}\right)^{\cos^2 x}$

Putting $\frac{1}{\cos^2 x}=y$, we get,

$L=\lim\limits_{y\to\infty}(1^y+2^y+\ldots\ldots\ldots+10^y)^{\frac{1}{y}}$

$= \lim\limits_{y\to\infty} 10\left[\left(\frac{1}{10}\right)^y + \left(\frac{2}{10}\right)^y + \ldots\ldots\ldots + \left(\frac{10}{10}\right)^y\right]^{\frac{1}{y}}$

$= 10(0 + 0 + \ldots\ldots\ldots + 1)^0 = 10$

Q85

$a_k = 2a_{k-1} - a_{k-2}$

$\Rightarrow a_1, a_2, \ldots, a_{11}$ are in AP with let common difference be d

$\therefore \quad \frac{a_1^2 + a_2^2 + \ldots + a_{11}^2}{11} = \frac{11a^2 + 35 \times 11\mathrm{d}^2 + 110a\mathrm{d}}{11} = 90$

$\Rightarrow 225 + 35\mathrm{d}^2 + 150\mathrm{d} = 90$

$35\mathrm{d}^2 + 150\mathrm{d} + 135 = 0$

$\Rightarrow \mathrm{d} = -3, -\frac{9}{7}$

Given, $a_2 < \frac{27}{2} \therefore \mathrm{d} = -3$ and $\mathrm{d} \neq -\frac{9}{7}$

Hence, $a_1 + a_2 + \ldots\ldots\ldots a_{11} = \frac{11}{2}[2 \times 15 + 10(-3)] = 0$

Q86

For non-trivial solutions,

$$\begin{vmatrix} a & 1 & b \\ b & 1 & a \\ a & b & ab \end{vmatrix} = 0$$

$\Rightarrow (a-b)\left(a-b^2\right) = 0$

$\Rightarrow$ Either $a = b$ or $a = b^2$

When $a = b$, then ordered pairs are $(0,0),(1,1),(2,2),(3,3),(4,4)$

When $a = b^2$, then $(4, 2)$

Hence, number of ordered pairs are 6

Q87

$\det.(\mathrm{adj}(\mathrm{adj}(\mathrm{adj}(\mathrm{adjA})))) = |\mathrm{A}|^{(3-1)^4}$

$= |\mathrm{A}|^{16} = 4^8 . 5^{16}$

$\Rightarrow |\mathrm{A}| = \pm 10$

$$|\mathrm{A}| = \begin{vmatrix} \mathrm{x} & \mathrm{y} & -\mathrm{z} \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{vmatrix} = \mathrm{x} + \mathrm{y} + \mathrm{z} = \pm 10$$

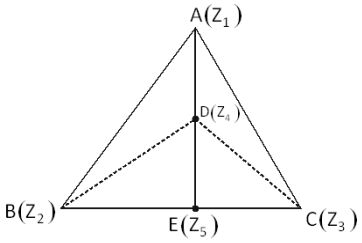
$\because \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathrm{N} \Rightarrow \mathrm{x} + \mathrm{y} + \mathrm{z} = -10$ (not possible)

Hence, $\mathrm{x} + \mathrm{y} + \mathrm{z} = 10$

The number of such matrices $= {}^{9}C_2$

$= 36$

Q88



Here, $Z_4 = \frac{\left(Z_1 + \frac{Z_2+Z_3}{2}\right)}{2}$

Let, midpoint of the line joining B & C is $E(Z_5)$, where $Z_5 = \frac{Z_2+Z_3}{2}$, then

Z_4 is midpoint of Z_1 & Z_5

$\Rightarrow \frac{\text{Area of } \Delta BDC}{\text{Area of } \Delta BAC} = \frac{1}{2} = 0.5$

Q89

$f(xf(y)) = x^p y^4$, put $x = \frac{1}{f(y)}$

$\therefore \quad f(1) = \left(\frac{1}{f(y)}\right)^p y^4 = \frac{y^4}{(f(y))^p}$

For $\quad y = 1, \quad f(1) = \frac{1}{(f(1))^p} \Rightarrow f(1) = 1$

$\therefore \quad f(y) = y^{4/p}$ ...(1)

$\therefore \quad f\left(xy^{4/p}\right) = x^p y^4$

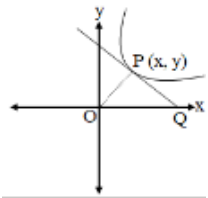
Put $\quad y = z^{p/4}$

$\therefore \quad f(xz) = x^p z^p \Rightarrow f(x) = x^p$ ...(2)

$\frac{4}{p} = p \quad \Rightarrow \quad p = 2$

Q90

(Y–y)= $\frac{dy}{dx}$ (X–x)



Thus meet x axis at $\left(x - y\frac{dy}{dx}, 0\right)$

OP=OQ $\sqrt{x^2+y^2}$=x–y $\frac{dy}{dx}$

$-\frac{ydx+xdy}{\sqrt{x^2+y^2}}$=dy

$\frac{ydx-xdy}{y^2\sqrt{1+x^2/y^2}} = \frac{-1}{y}$ dy

$\frac{1}{\sqrt{1+\left(\frac{x}{y}\right)^2}}d\left(\frac{x}{y}\right)$=– $\frac{1}{y}$ dy

$\log\left\{\frac{x}{y}+\sqrt{1+\left(\frac{x}{y}\right)^2}\right\}$=– log y+log k

$\frac{x}{y}+\sqrt{1+\frac{x^2}{y^2}}=\frac{k}{y}$

$x+\sqrt{x^2+y^2}$=k

passes through (1,0)

k=2

$x^2+y^2=(2-x)^2$

$y^2=4-4x$

$y^2=-4(x-1)$

vertex (1, 0)

tangent at vertex

x=1 $\Rightarrow$a=1